



# Ego-network data, composition, structure

## **Social Network Analysis**

ESOL • AY 2025-26

Raffaele Vacca

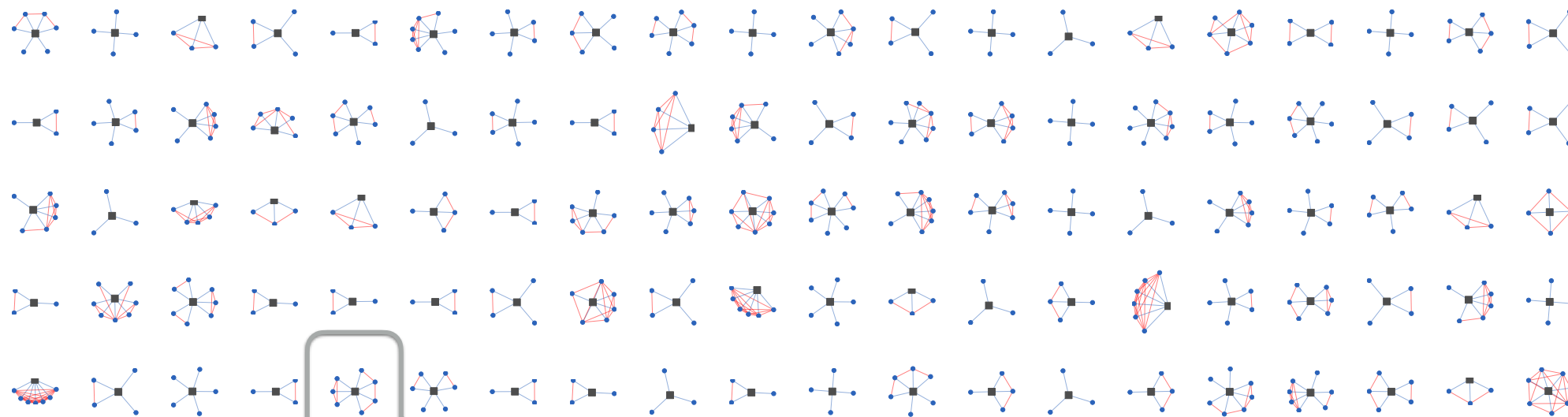
Department of social and political sciences  
University of Milan

# Outline

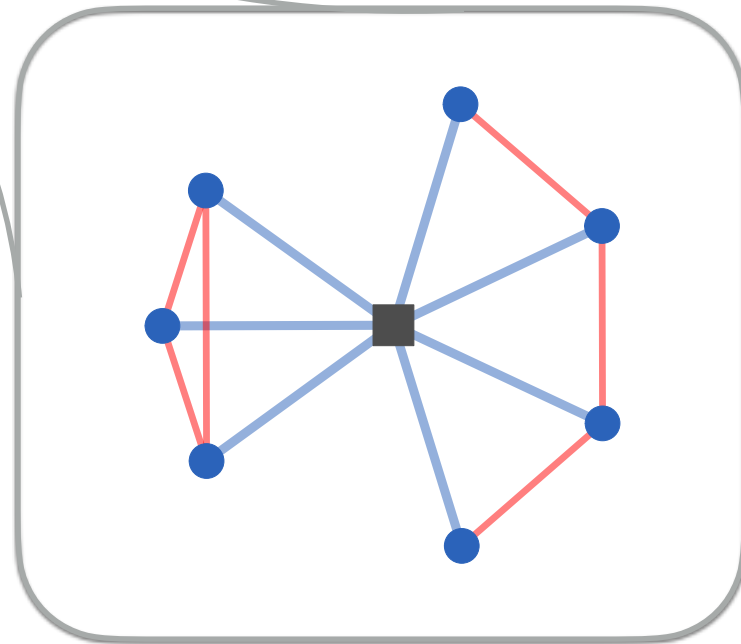
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1. The egocentric approach
2. Egocentric research design
3. Surveys on personal networks
- 4. Ego-network data**
5. Ego-network composition
6. Ego-network structure
7. Multilevel models for egocentric networks

# Ego-network data

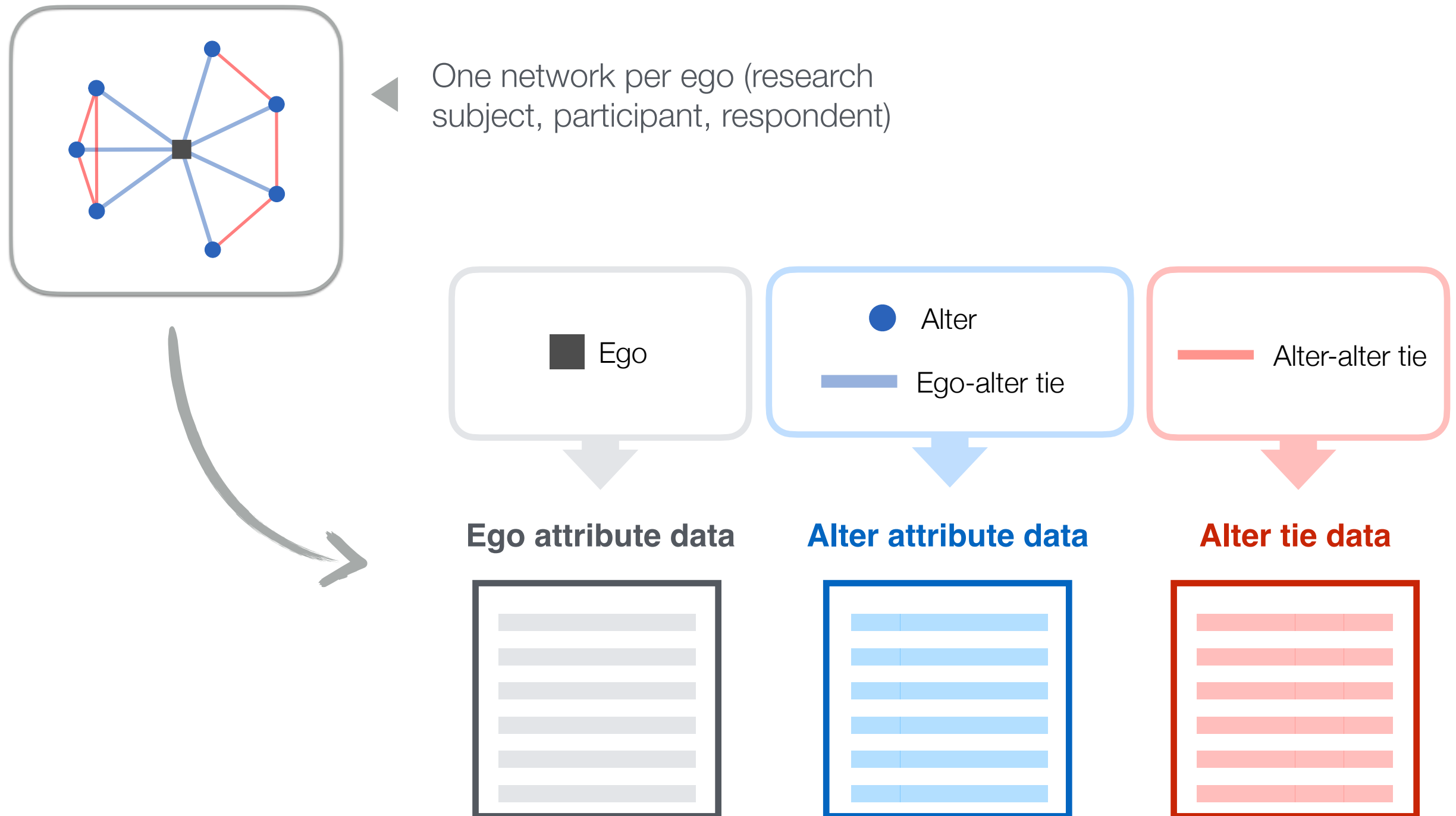


◀ All ego-networks  
in the data

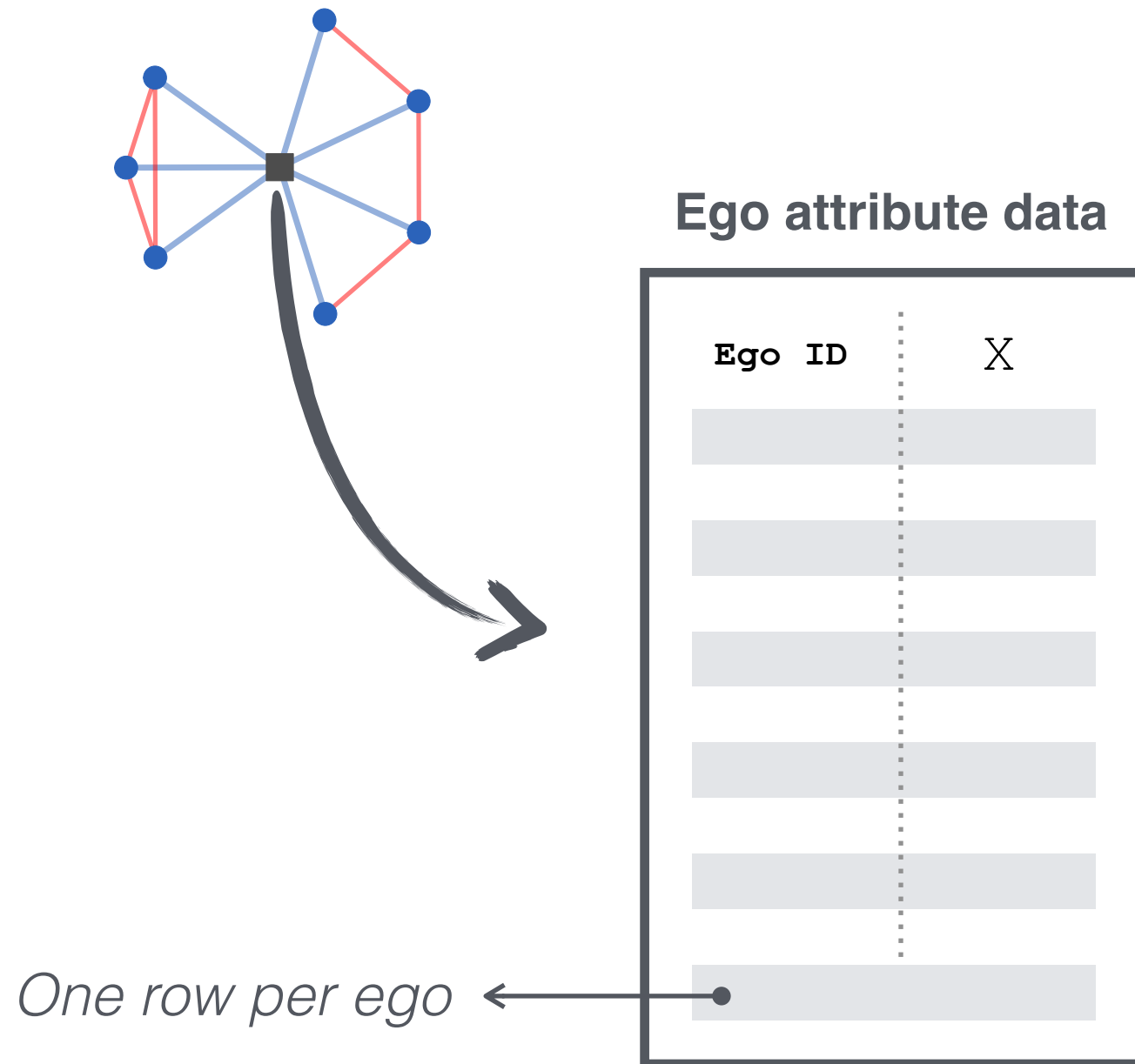


◀ One network per ego (research  
subject, participant, respondent)

# Ego-network data: contents and ideal format

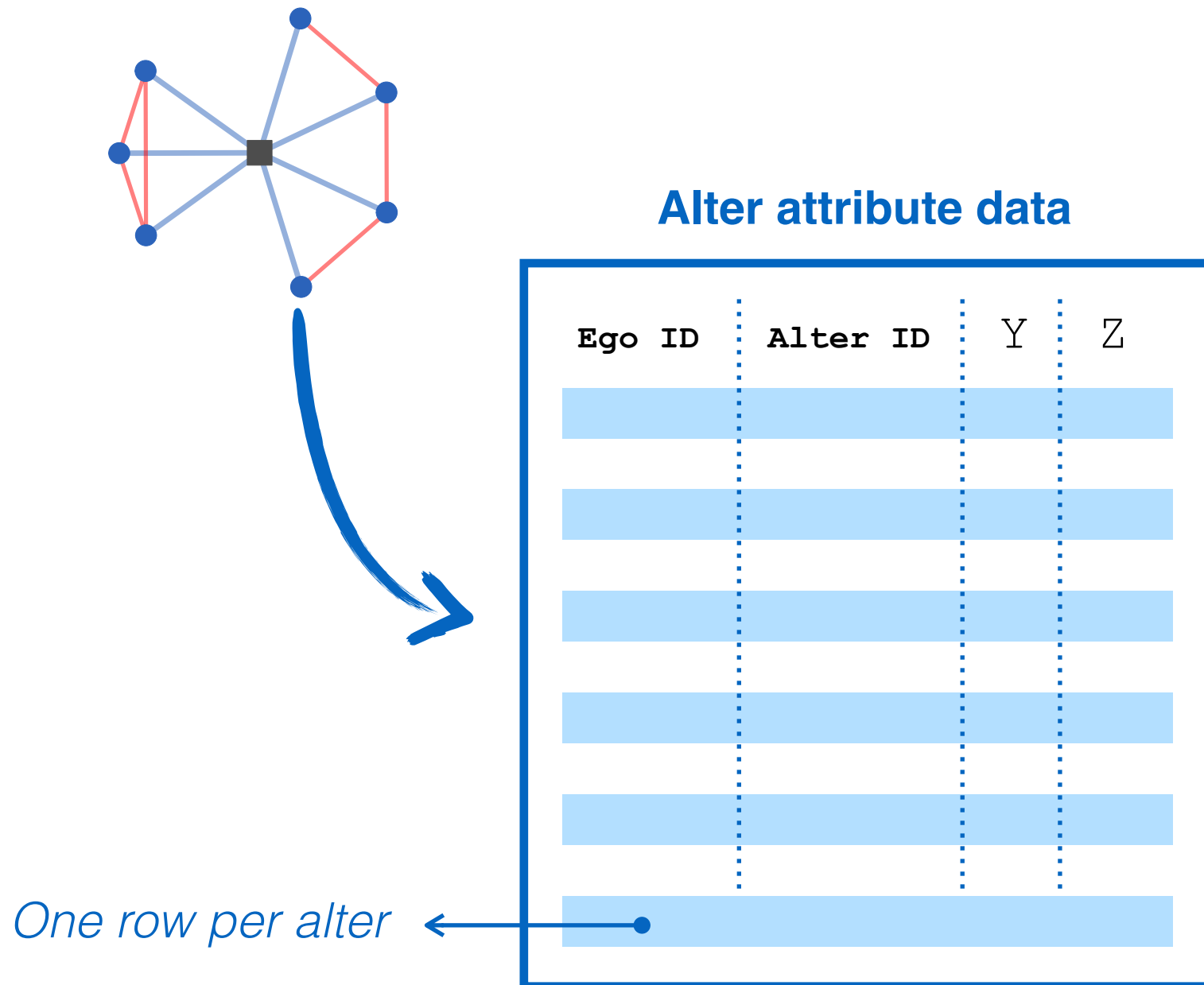


# Ego-network data: contents and ideal format



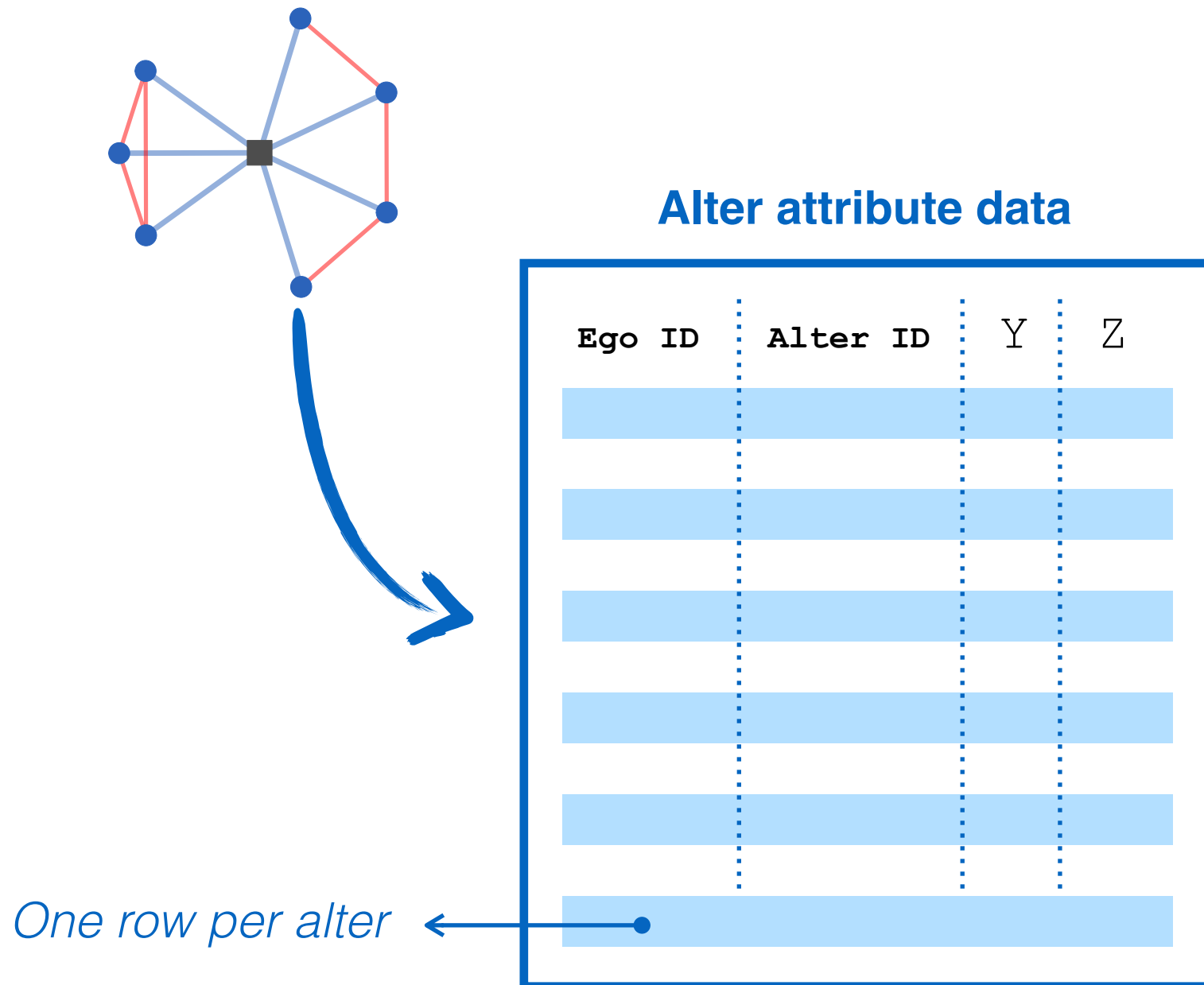
- **X**: Attributes of egos
- From standard questions about **respondent**
- **Ego level**: Variable has one value for each ego
- **Examples**:
  - Standard socio-demographics
  - Dependent variables (e.g. acculturation score, depression measure, smoking behavior)
  - Control variables relevant to your study (e.g. time since migration, health history)

# Ego-network data: contents and ideal format



- **Y**: Attributes of **alters**
- **Z**: Characteristics of **ego-alter ties**
- From **name interpreters**
- **Alter level**: Variable has one value for each alter *or* for each ego-alter tie
- One-to-one **correspondence** between alters and ties
  - Alter level and tie level are the same

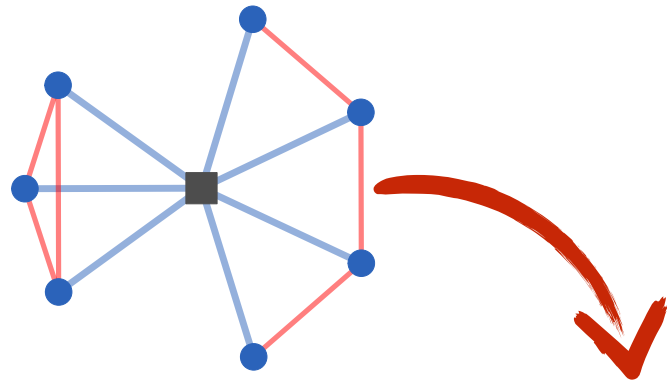
# Ego-network data: contents and ideal format



- **Examples:**

- Alters' gender, age bracket, ...
- Alter variables that are relevant to ego outcome: alter race/ethnicity, support provided to ego, smoking behavior, ...
- Type of relationship (e.g. close family, extended family, friend, acquaintance, ...)
- Focus/context of socialization (e.g., family, school, work, neighborhood)
- Measures of tie strength, trust, frequency of contact, ...

# Ego-network data: contents and ideal format



Alter tie data

| Ego ID | Sending alter ID | Receiving alter ID | W |
|--------|------------------|--------------------|---|
|        |                  |                    |   |
|        |                  |                    |   |
|        |                  |                    |   |
|        |                  |                    |   |
|        |                  |                    |   |
|        |                  |                    |   |

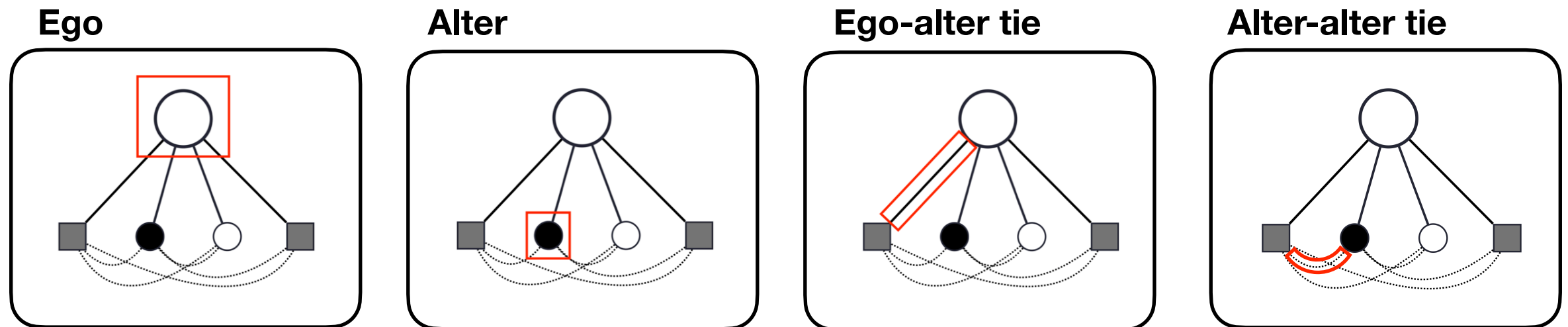
One row per  
alter-alter tie

- **W**: Characteristics of **alter-alter ties**
- From **edge interpreters**
- Typically undirected ties (reported by ego)
- Examples:
  - Whether alters A and B know each other
  - How close alters A and B are

# Ego-network data: contents and ideal format

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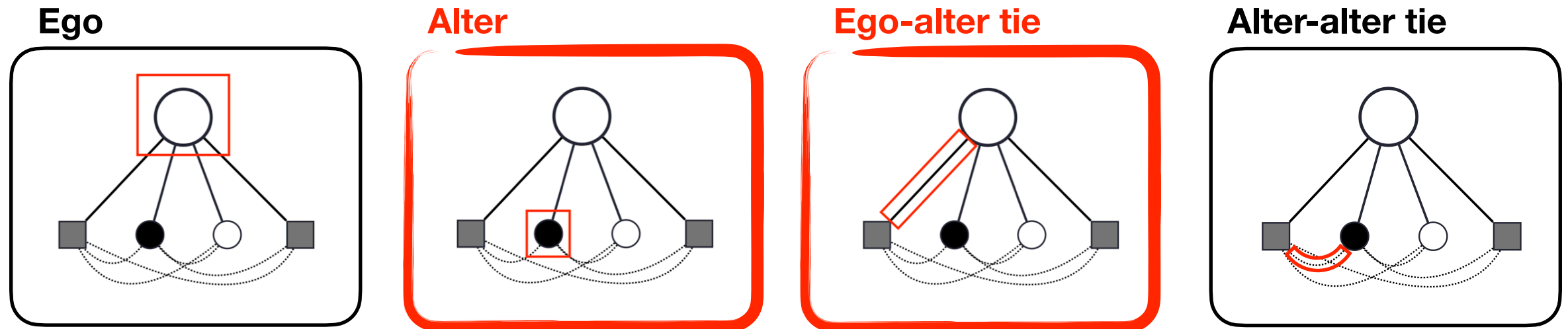
- So in ego-network data variables can be at any of these **levels**:



# Ego-network data: contents and ideal format

---

- So in ego-network data variables can be at any of these **levels**:



*Same level in typical ego-network data*

# 1. Ego attributes

---

- **Ego-level data** upon data import, before analysis

| ego_ID | age | sex | accult_score |
|--------|-----|-----|--------------|
| 5      | 61  | M   | 4.33         |
| 20     | 43  | F   | 2.2          |
| 31     | 39  | F   | 1            |
| 40     | 25  | M   | 3.39         |

# 1. Ego attributes

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Data units

# 1. Ego attributes

---

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| ego_ID | age | sex | accult_score |
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| 31     | 39  | F   | 1            |
| 40     | 25  | M   | 3.39         |

Non-network ego-level attributes

## 2. Alter attributes

---

- **Alter-level data** upon data import, before analysis

| alter_ID | ego_ID | alter_sex | alter_age | alter_trust | alter_nat |
|----------|--------|-----------|-----------|-------------|-----------|
| 501      | 5      | F         | 21–30     | 4           | slk       |
| 502      | 5      | F         | 31–40     | 4           | oth       |
| 503      | 5      | M         | 21–30     | 2           | ita       |
| 504      | 5      | F         | >60       | 5           | sly       |
| 505      | 5      | F         | 41–50     | 2           | slk       |
| 2001     | 20     | M         | 31–40     | 2           | ita       |
| 2002     | 20     | F         | 15–20     | 3           | slk       |
| 2003     | 20     | M         | 21–30     | 4           | slk       |
| 3101     | 31     | F         | 41–50     | 1           | ita       |
| 3102     | 31     | F         | >60       | 3           | slk       |

## 2. Alter attributes

---

- **Alter-level data** upon data import, before analysis

| alter_ID | ego_ID | alter_sex | alter_age | alter_trust | alter_nat |
|----------|--------|-----------|-----------|-------------|-----------|
| 501      | 5      | F         | 21-30     | 4           | slk       |
| 502      | 5      | F         | 31-40     | 4           | oth       |
| 503      | 5      | M         | 21-30     | 2           | ita       |
| 504      | 5      | F         | >60       | 5           | sly       |
| 505      | 5      | F         | 41-50     | 2           | slk       |
| 2001     | 20     | M         | 31-40     | 2           | ita       |
| 2002     | 20     | F         | 15-20     | 3           | slk       |
| 2003     | 20     | M         | 21-30     | 4           | slk       |
| 3101     | 31     | F         | 41-50     | 1           | ita       |
| 3102     | 31     | F         | >60       | 3           | slk       |

Data units

## 2. Alter attributes

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- **Alter-level data** upon data import, before analysis

| alter_ID | ego_ID | alter_sex | alter_age | alter_trust | alter_nat |
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| 503      | 5      | M         | 21-30     | 2           | ita       |
| 504      | 5      | F         | >60       | 5           | sly       |
| 505      | 5      | F         | 41-50     | 2           | slk       |
| 2001     | 20     | M         | 31-40     | 2           | ita       |
| 2002     | 20     | F         | 15-20     | 3           | slk       |
| 2003     | 20     | M         | 21-30     | 4           | slk       |
| 3101     | 31     | F         | 41-50     | 1           | ita       |
| 3102     | 31     | F         | >60       | 3           | slk       |

Ego who nominated each alter

## 2. Alter attributes

---

- **Alter-level data** upon data import, before analysis

| alter_ID | ego_ID | alter_sex | alter_age | alter_trust | alter_nat |
|----------|--------|-----------|-----------|-------------|-----------|
| 501      | 5      | F         | 21-30     | 4           | slk       |
| 502      | 5      | F         | 31-40     | 4           | oth       |
| 503      | 5      | M         | 21-30     | 2           | ita       |
| 504      | 5      | F         | >60       | 5           | sly       |
| 505      | 5      | F         | 41-50     | 2           | slk       |
| 2001     | 20     | M         | 31-40     | 2           | ita       |
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| 2003     | 20     | M         | 21-30     | 4           | slk       |
| 3101     | 31     | F         | 41-50     | 1           | ita       |
| 3102     | 31     | F         | >60       | 3           | slk       |

Non-network alter-level attributes

### 3. Alter-alter ties

---

- **Alter-alter tie data** upon data import, before analysis

ego\_ID=5

| 5   | 501 | 502 | 503 |
|-----|-----|-----|-----|
| 501 | NA  | 1   | 0   |
| 502 | 1   | NA  | 1   |
| 503 | 0   | 1   | NA  |

ego\_ID=20

| 50   | 2001 | 2002 | 2003 |
|------|------|------|------|
| 2001 | NA   | 1    | 1    |
| 2002 | 1    | NA   | 1    |
| 2003 | 1    | 1    | NA   |

ego\_ID=31

| 31   | 3101 | 3102 | 3103 |
|------|------|------|------|
| 3101 | NA   | 0    | 0    |
| 3102 | 0    | NA   | 1    |
| 3103 | 0    | 1    | NA   |

ego\_ID=40

| 40   | 4001 | 4002 | 4003 |
|------|------|------|------|
| 4001 | NA   | 1    | 0    |
| 4002 | 1    | NA   | 0    |
| 4003 | 1    | 0    | NA   |

(Network data as **adjacency matrices**)

### 3. Alter-alter ties

---

- **Alter-alter tie data** upon data import, before analysis

ego\_ID=5

| ego_ID | from | to  |
|--------|------|-----|
| 5      | 501  | 502 |
| 5      | 502  | 503 |
| 5      | 501  | 515 |

ego\_ID=20

| ego_ID | from | to   |
|--------|------|------|
| 20     | 2001 | 2002 |
| 20     | 2002 | 2003 |
| 20     | 2001 | 2003 |

ego\_ID=31

| ego_ID | from | to   |
|--------|------|------|
| 31     | 3101 | 3104 |
| 31     | 3102 | 3110 |
| 31     | 3102 | 3115 |

ego\_ID=40

| ego_ID | from | to   |
|--------|------|------|
| 40     | 4001 | 4002 |
| 40     | 4001 | 4003 |
| 40     | 4002 | 4005 |

(Network data as **multiple edge lists**)

### 3. Alter-alter ties

---

- **Alter-alter tie data** upon data import, before analysis

| ego_ID | from | to   |           |
|--------|------|------|-----------|
| 5      | 501  | 502  | ego_ID=5  |
| 5      | 502  | 503  |           |
| 5      | 501  | 515  |           |
| 20     | 2001 | 2002 | ego_ID=20 |
| 20     | 2002 | 2003 |           |
| 20     | 2001 | 2003 |           |
| 31     | 3101 | 3104 | ego_ID=31 |
| 31     | 3102 | 3110 |           |
| 31     | 3102 | 3115 |           |

(Network data as **single edge list**)

# Alter-level data

---

- This is called **level 1** in multilevel analysis

| <b>alter_ID</b> | <b>ego_ID</b> | <b>alter_sex</b> | <b>alter_age</b> | <b>alter_trust</b> | <b>alter_nat</b> |
|-----------------|---------------|------------------|------------------|--------------------|------------------|
| 501             | 5             | F                | 21–30            | 4                  | slk              |
| 502             | 5             | F                | 31–40            | 4                  | oth              |
| 503             | 5             | M                | 21–30            | 2                  | ita              |
| 504             | 5             | F                | >60              | 5                  | sly              |
| 505             | 5             | F                | 41–50            | 2                  | slk              |
| 2001            | 20            | M                | 31–40            | 2                  | ita              |
| 2002            | 20            | F                | 15–20            | 3                  | slk              |
| 2003            | 20            | M                | 21–30            | 4                  | slk              |
| 3101            | 31            | F                | 41–50            | 1                  | ita              |
| 3102            | 31            | F                | >60              | 3                  | slk              |

# Alter-level data

---

- This is called **level 2** in multilevel analysis

| alter_ID | ego_ID | alter_sex | alter_age | alter_trust | alter_nat |
|----------|--------|-----------|-----------|-------------|-----------|
| 501      | 5      | F         | 21-30     | 4           | slk       |
| 502      | 5      | F         | 31-40     | 4           | oth       |
| 503      | 5      | M         | 21-30     | 2           | ita       |
| 504      | 5      | F         | >60       | 5           | sly       |
| 505      | 5      | F         | 41-50     | 2           | slk       |
| 2001     | 20     | M         | 31-40     | 2           | ita       |
| 2002     | 20     | F         | 15-20     | 3           | slk       |
| 2003     | 20     | M         | 21-30     | 4           | slk       |
| 3101     | 31     | F         | 41-50     | 1           | ita       |
| 3102     | 31     | F         | >60       | 3           | slk       |

# Ego-level data

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- This is called **level 2** in multilevel analysis

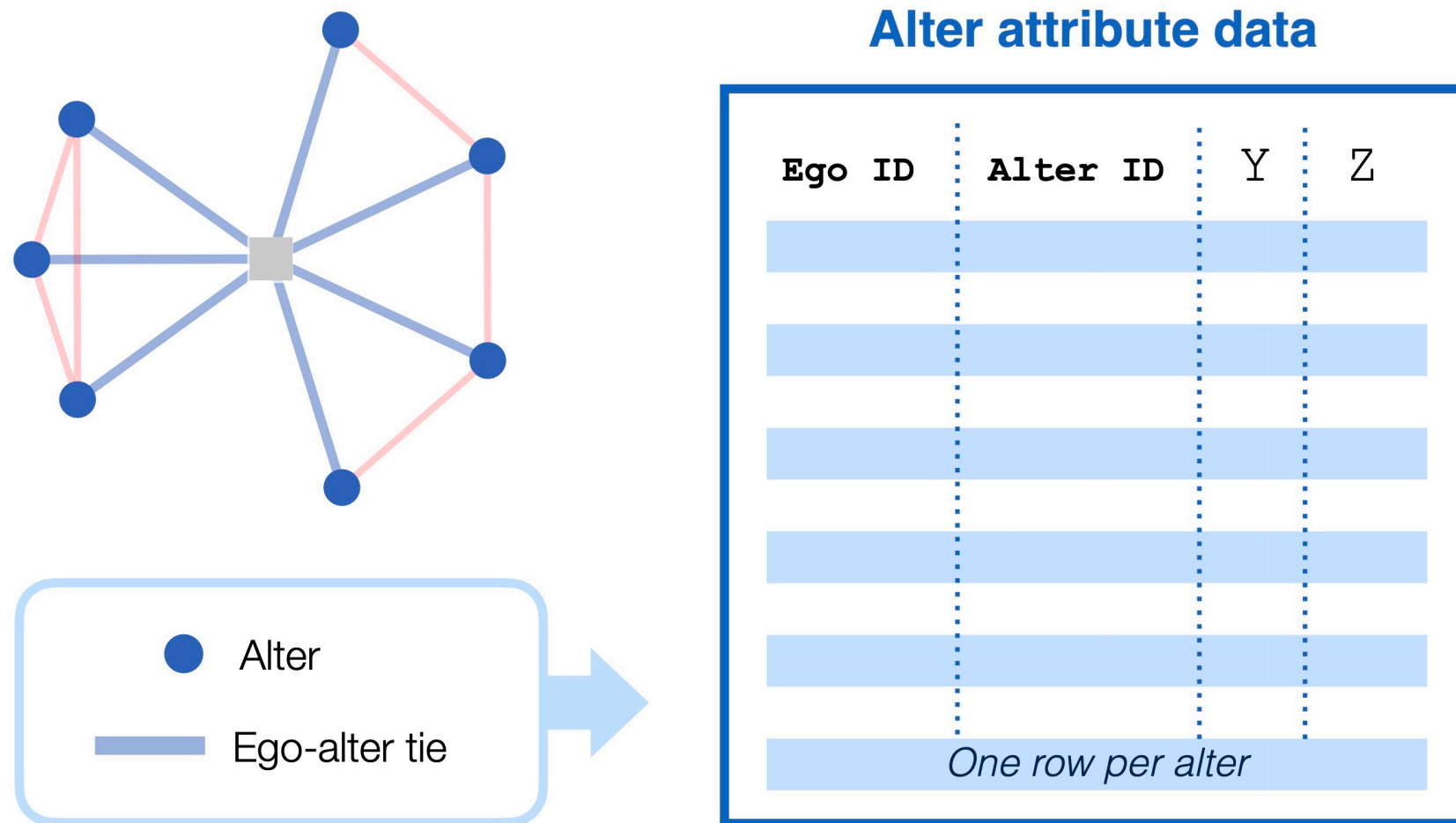
| <b>ego_ID</b> | <b>age</b> | <b>sex</b> | <b>accult_score</b> |
|---------------|------------|------------|---------------------|
| 5             | 61         | M          | 4.33                |
| 20            | 43         | F          | 2.2                 |
| 31            | 39         | F          | 1                   |
| 40            | 25         | M          | 3.39                |

# Outline

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1. The egocentric approach
2. Egocentric research design
3. Surveys on personal networks
4. Ego-network data
- 5. Ego-network composition**
6. Ego-network structure
7. Multilevel models for egocentric networks

# Ego-network composition



- Ego-network **composition** is the distribution of **attributes of alters or ego-alter ties**
- Different attribute distributions create different social worlds for the ego in terms of **socio-demographics, behaviors, social norms**, etc.

# Examples of attributes

---

|                    | Attribute of alters (Y)         | Characteristic of ego-alter ties (Z)                        |
|--------------------|---------------------------------|-------------------------------------------------------------|
| <b>Categorical</b> | Gender, Race/ethnicity of alter | Context of relationship, Frequency of ego-alter contact     |
| <b>Continuous</b>  | Age of alter                    | N ego-alter contacts last week, N years ego has known alter |

- Ego-network **composition** is the distribution of **attributes of alters or ego-alter ties**
- Different attribute distributions create different social worlds for the ego in terms of **socio-demographics, behaviors, social norms**, etc.

# Distributions of interest

- Compositional analyses may be interested in one of **3 distributions**:
  1. Distribution of an **alter/tie attribute** in **one** personal network
    - Analysis results in **ego-level summary** measure (e.g. average alter age for one ego)
  2. Distribution of an **alter/tie attribute** across **many/all** personal networks
    - E.g. average alter age among all alters in the data
  3. Distribution of an ego-level **summary measure** from analysis (1) among **many/all** personal networks
    - E.g. Mean and SD of ego-level average alter age among egos

|                   |                          | How many networks considered |      |
|-------------------|--------------------------|------------------------------|------|
|                   |                          | One                          | Many |
| Units of analysis | Alters or Ego-alter ties | 1                            | 2    |
|                   | Egos or Networks         | —                            | 3    |

# Ego-level summary measures

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- In many ego-network studies, the egos are the **units of analysis**
- This type of studies focus especially on **distributions (1) and (3)**

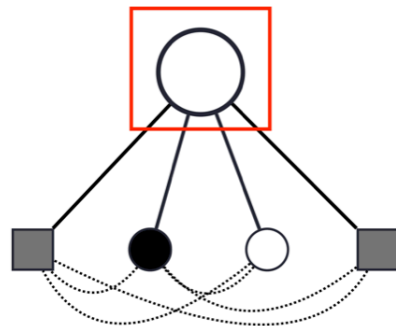
|                   |                          | How many networks considered |      |
|-------------------|--------------------------|------------------------------|------|
|                   |                          | One                          | Many |
| Units of analysis | Alters or Ego-alter ties | 1                            | 2    |
|                   | Egos or Networks         | —                            | 3    |

# Ego-level variables

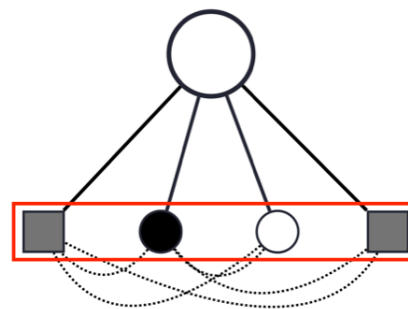
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- When the egos are the **units of analysis**, all variables need to be **ego-level**
  - I.e., with a value observed for each ego
- These ego-level variables can be of **three types**:

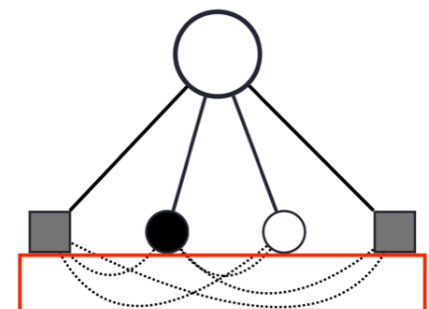
Characteristics of  
the Ego



Characteristics of  
ego-network composition



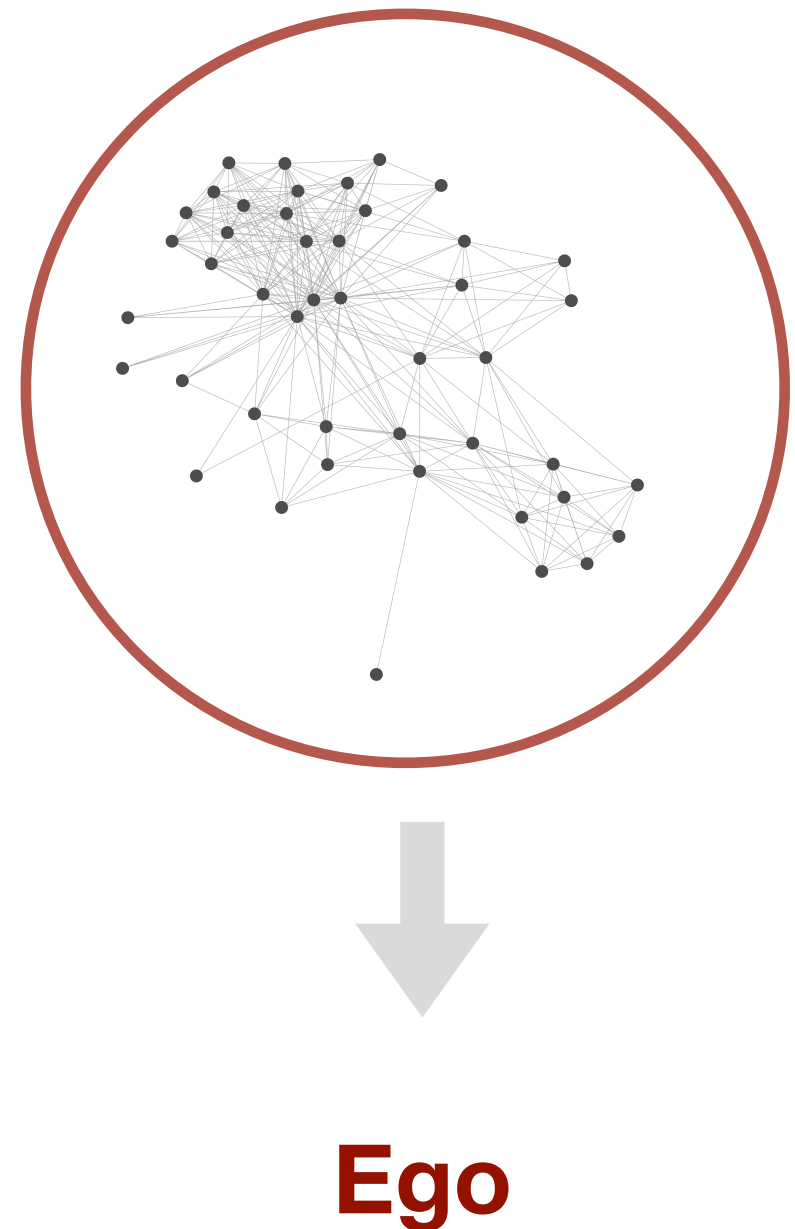
Characteristics of  
ego-network structure



# Ego-level summarization

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- This means that all measures based on characteristics of alters or ties (ego-alter or alter-alter ties) need to be **aggregated** or **summarized** at the ego level
- To summarize ego-network composition, **different aggregation/summarization functions** may be appropriate depending on
  - Measurement levels (nominal, interval, etc.)
  - Interest of analysis (e.g. prevalence of particular category or diversity?)



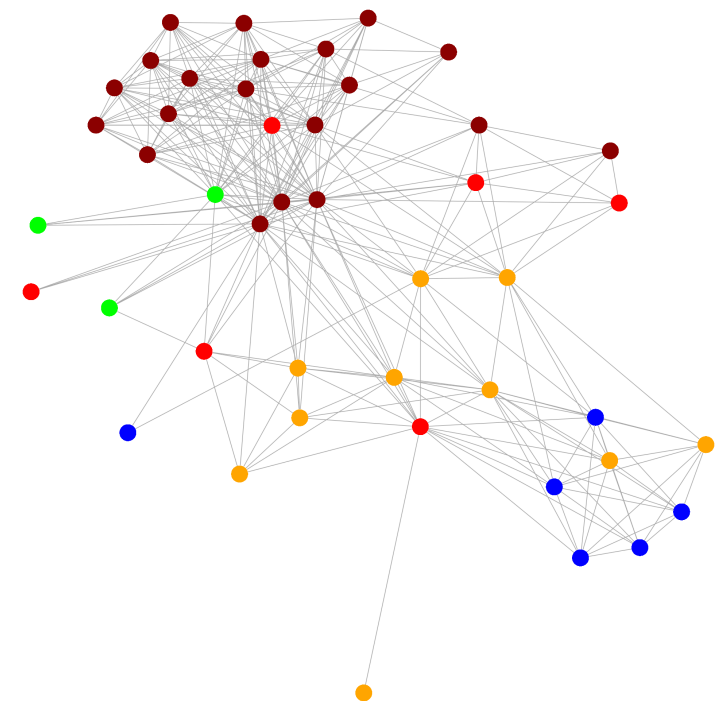
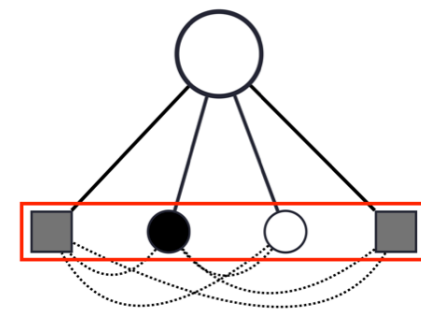
# Ego-level summarization of network composition

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- **Examples:**

- Alter gender → % women in personal network
- Alter age → Avg age in personal network
- Alter ethnicity → Ethnic diversity in personal network
- Strength of alter-ego tie → Avg tie strength in personal network

## Characteristics of ego-network composition



# Ego-level data *after* analysis

---

| ego_ID | age | sex | accult_score | prop_fem<br>ale | avg_alt_<br>age | density | N_compone<br>nts | max_alt_b<br>etweennes<br>s |
|--------|-----|-----|--------------|-----------------|-----------------|---------|------------------|-----------------------------|
| 5      | 61  | M   | 4.33         | 0.9             | 33              | 0.8     | 1                | 20                          |
| 20     | 43  | F   | 2.2          | 0.5             | 40              | 0.75    | 2                | 5                           |
| 31     | 39  | F   | 1            | 0.33            | 35              | 0.33    | 1                | 25                          |
| 40     | 25  | M   | 3.39         | 0.1             | 43              | 0.56    | 4                | 10                          |

Data units

# Ego-level data *after* analysis

---

| ego_ID | age | sex | accult_score | prop_fem<br>ale | avg_alt_<br>age | density | N_compone<br>nts | max_alt_b<br>etweennes<br>s |
|--------|-----|-----|--------------|-----------------|-----------------|---------|------------------|-----------------------------|
| 5      | 61  | M   | 4.33         | 0.9             | 33              | 0.8     | 1                | 20                          |
| 20     | 43  | F   | 2.2          | 0.5             | 40              | 0.75    | 2                | 5                           |
| 31     | 39  | F   | 1            | 0.33            | 35              | 0.33    | 1                | 25                          |
| 40     | 25  | M   | 3.39         | 0.1             | 43              | 0.56    | 4                | 10                          |

Non-network, ego-level attributes

# Ego-level data *after* analysis

---

| ego_ID | age | sex | accult_score | prop_fem<br>ale | avg_alt_<br>age | density | N_compone<br>nts | max_alt_b<br>etweennes<br>s |
|--------|-----|-----|--------------|-----------------|-----------------|---------|------------------|-----------------------------|
| 5      | 61  | M   | 4.33         | 0.9             | 33              | 0.8     | 1                | 20                          |
| 20     | 43  | F   | 2.2          | 0.5             | 40              | 0.75    | 2                | 5                           |
| 31     | 39  | F   | 1            | 0.33            | 35              | 0.33    | 1                | 25                          |
| 40     | 25  | M   | 3.39         | 0.1             | 43              | 0.56    | 4                | 10                          |

Ego-level compositional variables: summarization of ego-network composition

# What do we do with these summary measures?

---

- Ego-level summary measures of network composition are used in at least **3 ways**:
  - These are similar to the way we can use any variable in **social survey data**

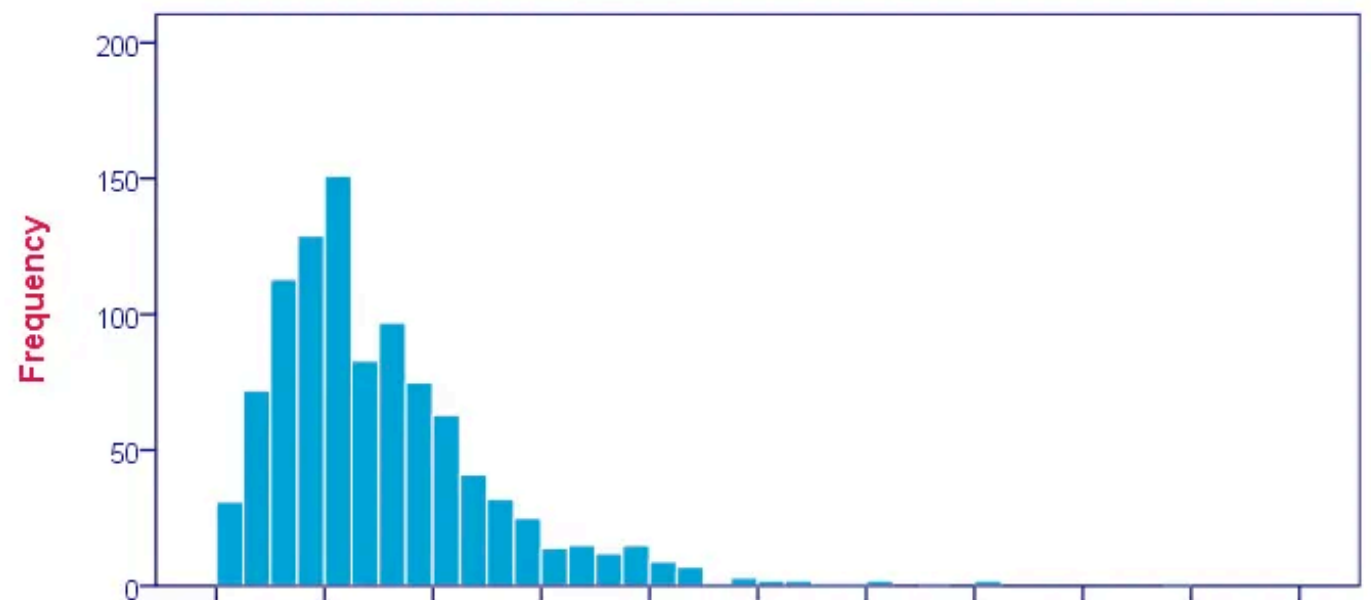
# What do we do with these summary measures?

---

- Ego-level summary measures of network composition are used in at least **3 ways**:

1. To describe the **univariate distribution** of network characteristics in our sample of ego (distribution 3 above)

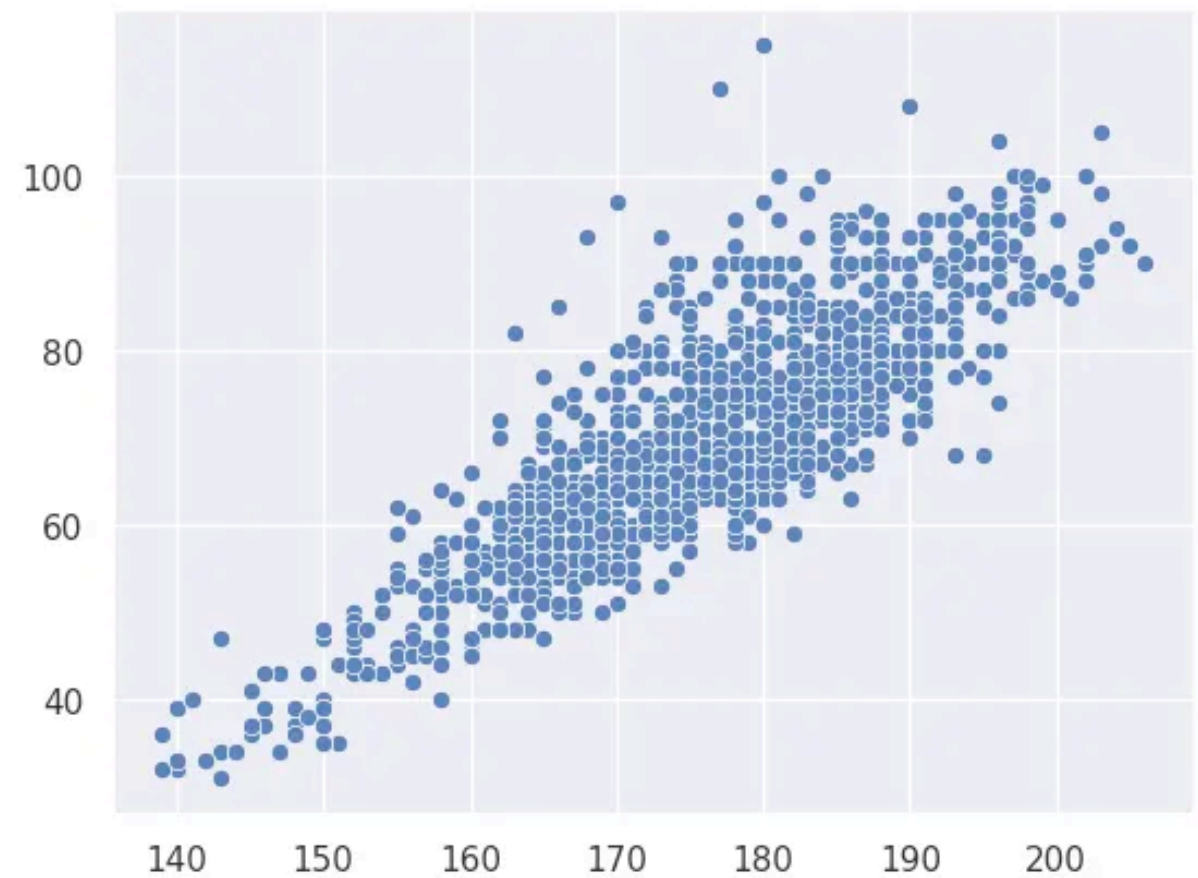
- E.g., what's the average alter age or the proportion of female alters in the average ego-network in our data?



# What do we do with these summary measures?

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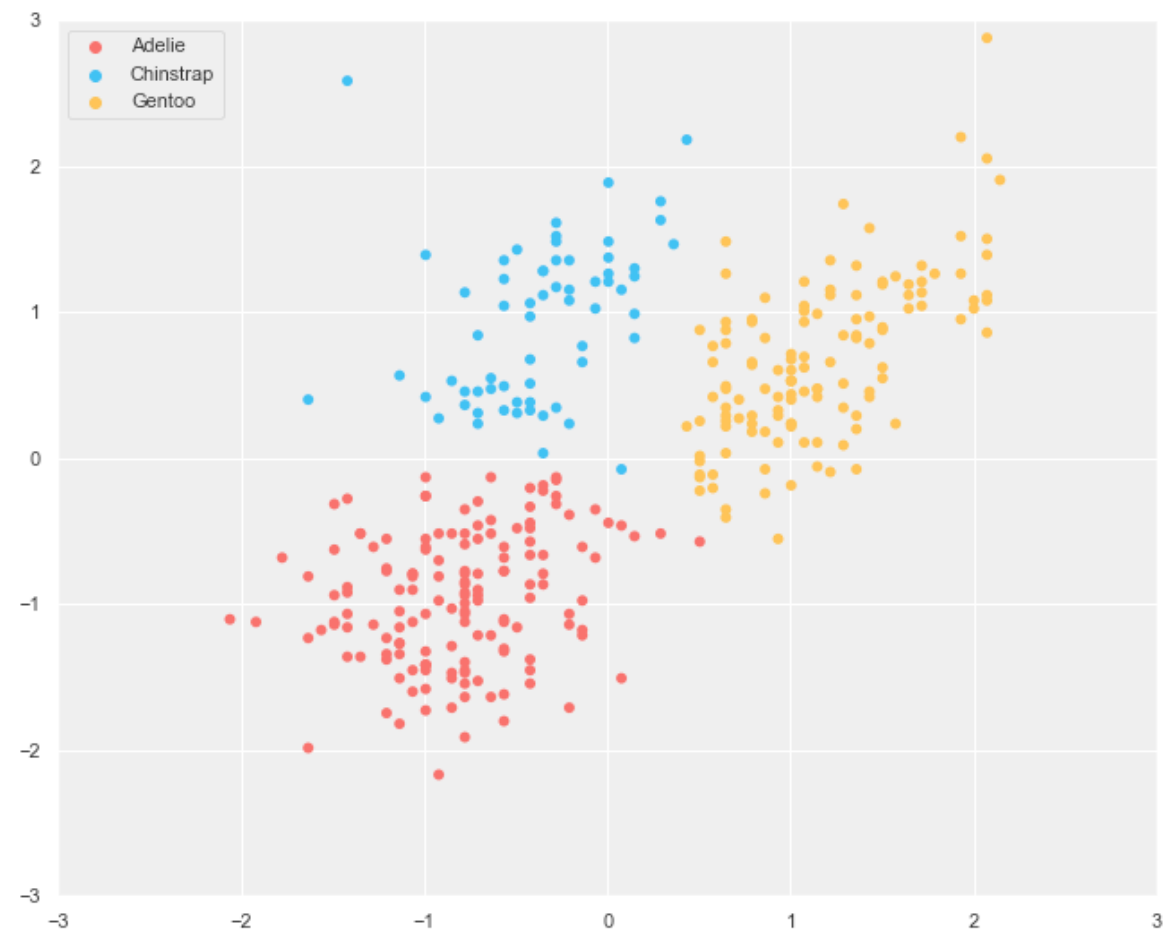
- Ego-level summary measures of network composition are used in at least **3 ways**:
  - To describe the **bivariate association** between ego-network characteristics or an ego-network characteristic and a non-network variable
    - E.g., how is the proportion of female alters in the ego-network correlated with ego's SES?



# What do we do with these summary measures?

---

- Ego-level summary measures of network composition are used in at least **3 ways**:
- Sometimes you do (1) or (2) with **cluster analysis**
  - Ego-network **typologies**

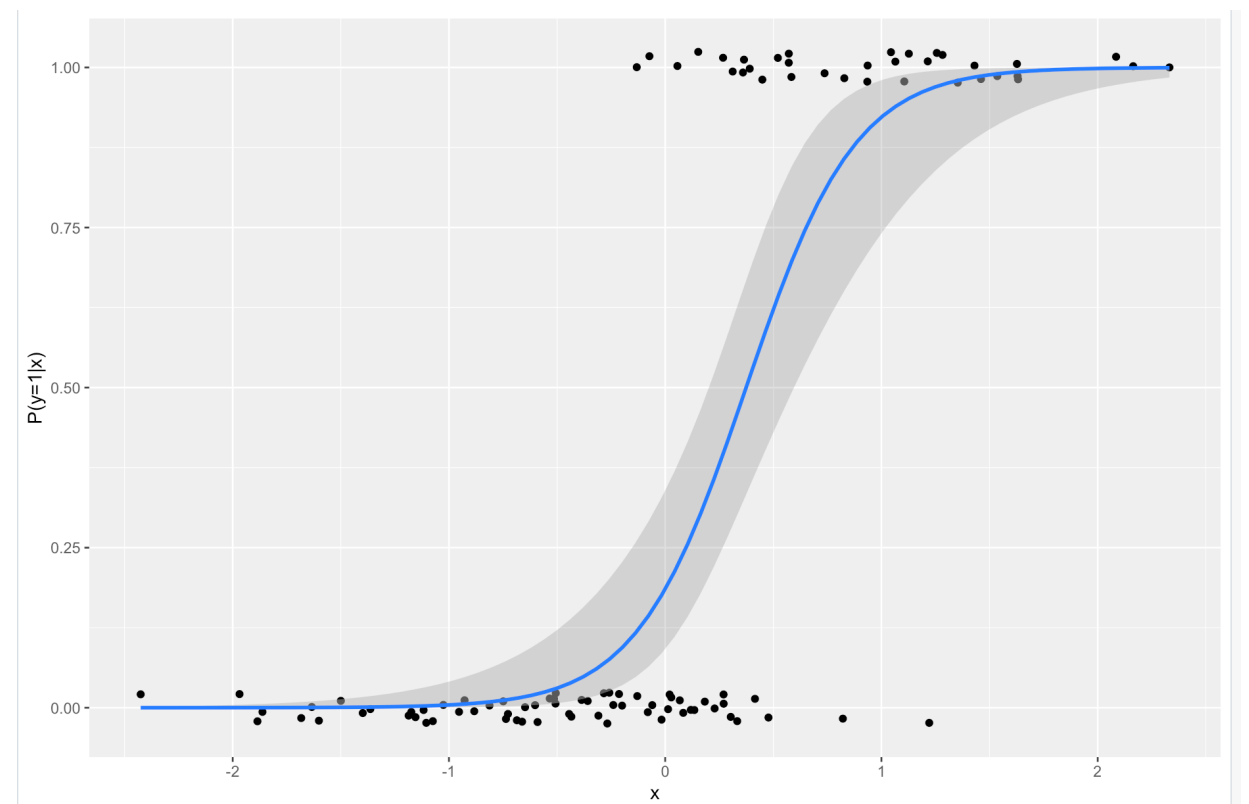


# What do we do with these summary measures?

- Ego-level summary measures of network composition are used in at least **3 ways**:

3. To estimate **statistical models** with

- Ego-network summary measure as **DV**:
  - I.e, we want to explain characteristics of personal networks
- Ego-network summary measure as **IV**
  - I.e, we want to explain non-network characteristics of the ego as a function of personal network features

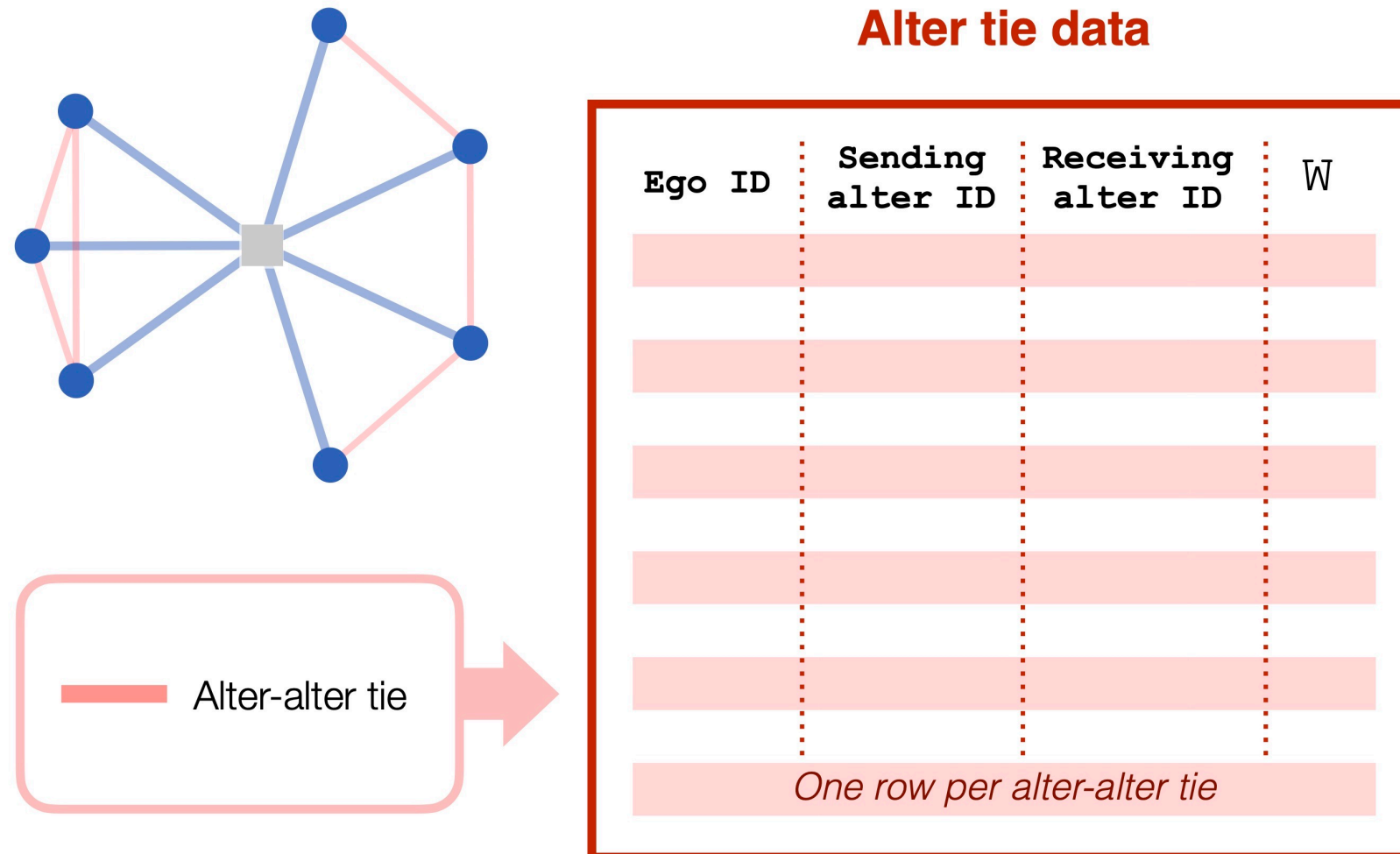


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# Ego-network structure



- Ego-network **structure** refers to the distribution of **alter-alter ties** around the ego
- Different distributions create different **structural positions and properties** for both the ego and the alters

# Examples of alter-alter tie variables

---

| Weight of alter-alter tie (W) |                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Dichotomous</b>            | <i>Do Alter 1 and Alter 2 know each other?</i><br>1 = Yes, 2 = Maybe                                            |
| <b>Polytomous</b>             | <i>How close are Alter 1 and Alter 2 from 1 to 3?</i><br>1 = A little close, 2 = Somewhat close, 3 = Very close |

---

- Ego-network **structure** refers to the distribution of **alter-alter ties** around the ego
- Different distributions create different **structural positions and properties** for both the ego and the alters

# Distributions of interest

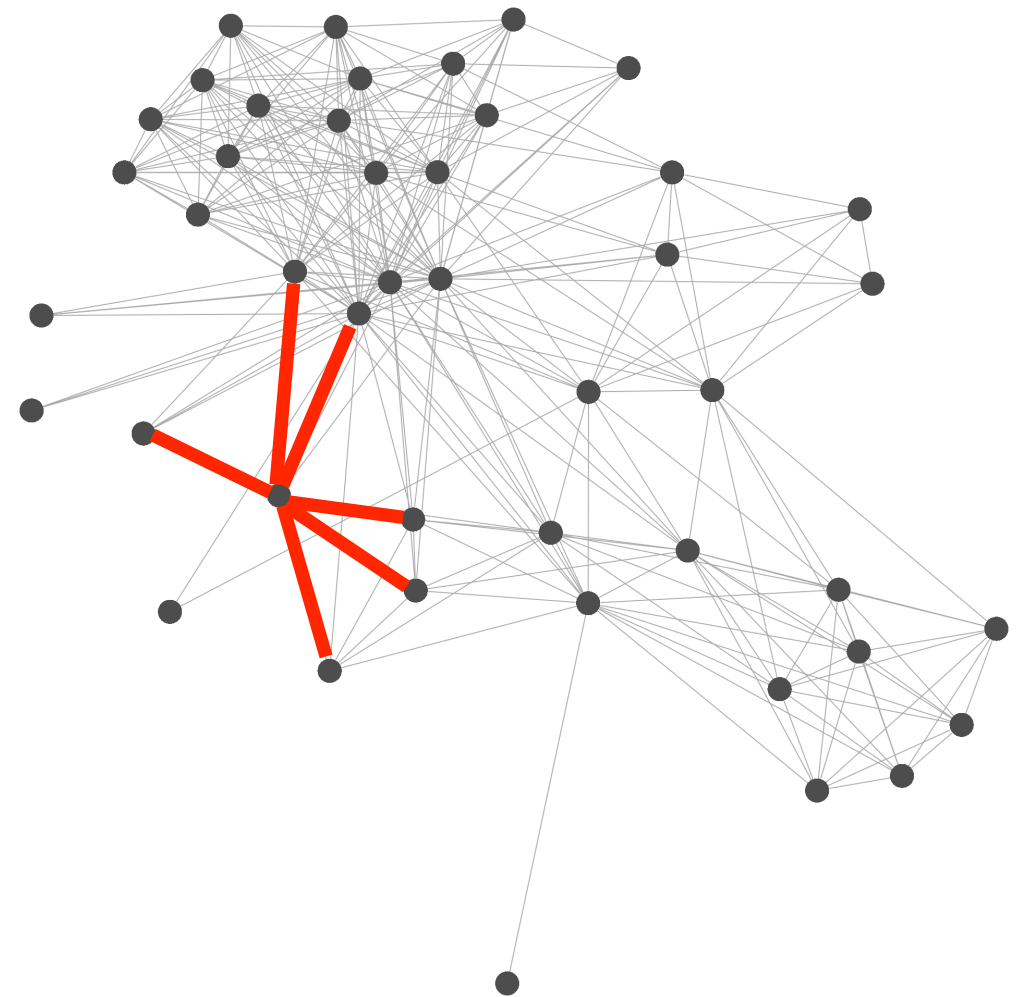
- Structural analyses may be interested in one of **3 distributions**:
  1. Distribution of a **structural characteristic** of alters or alter-alter ties in **one** personal network
    - Analysis results in ego-level summary measure (e.g. average alter degree or network density for one ego)
  2. Distribution of a **structural characteristic** of alters or alter-alter ties across **all alters** and networks in the data
    - E.g. average alter degree among all alters in the data
  3. Distribution of a **structural characteristic** of egos or networks **among the egos** in the data
    - E.g. Mean and SD of ego-network density among egos

|                   |                            | How many networks considered |      |
|-------------------|----------------------------|------------------------------|------|
|                   |                            | One                          | Many |
| Units of analysis | Alters or Alter-alter ties | 1                            | 2    |
|                   | Egos or Networks           | —                            | 3    |

# Ego-network structural measures

---

- Any **SNA structural measure** can be applied to a personal network to obtain
  - Node-level structural measures
  - Dyad-level structural measures
  - Network-level structural measures
- **Examples:**
  - Node-level: Alter betweenness centrality
  - Dyad-level: Geodesic distance between two alters (excluding ego node)
  - Network-level: Density of personal network

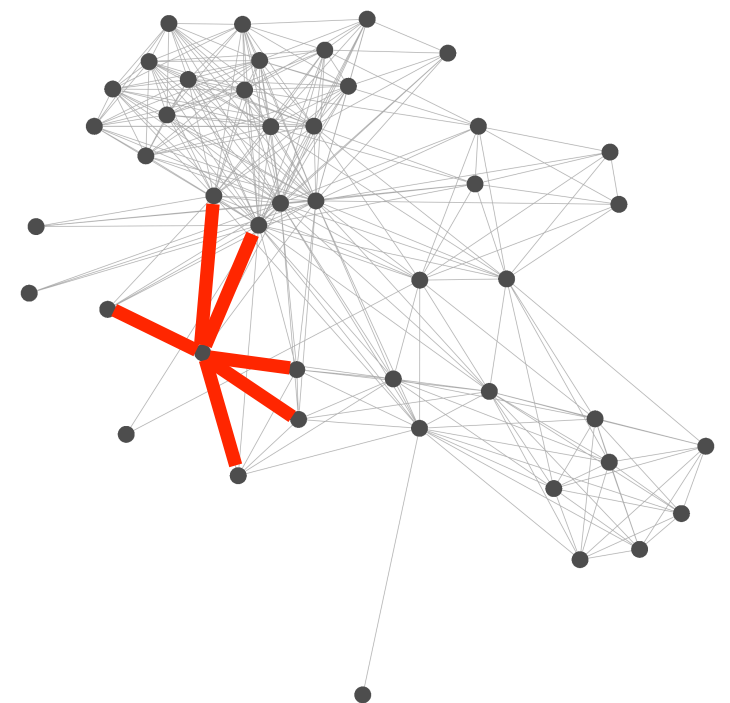
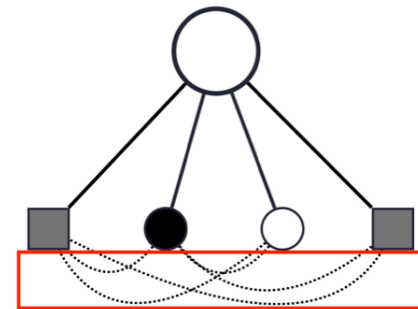


# Ego-network structural measures

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- In an ego-level analysis, these measures need to be **summarized for each ego** (i.e., ego-network), for example:
  - Max betweenness centrality of any alter
  - Avg geodesic distance in any pair of alters
  - Density of personal network

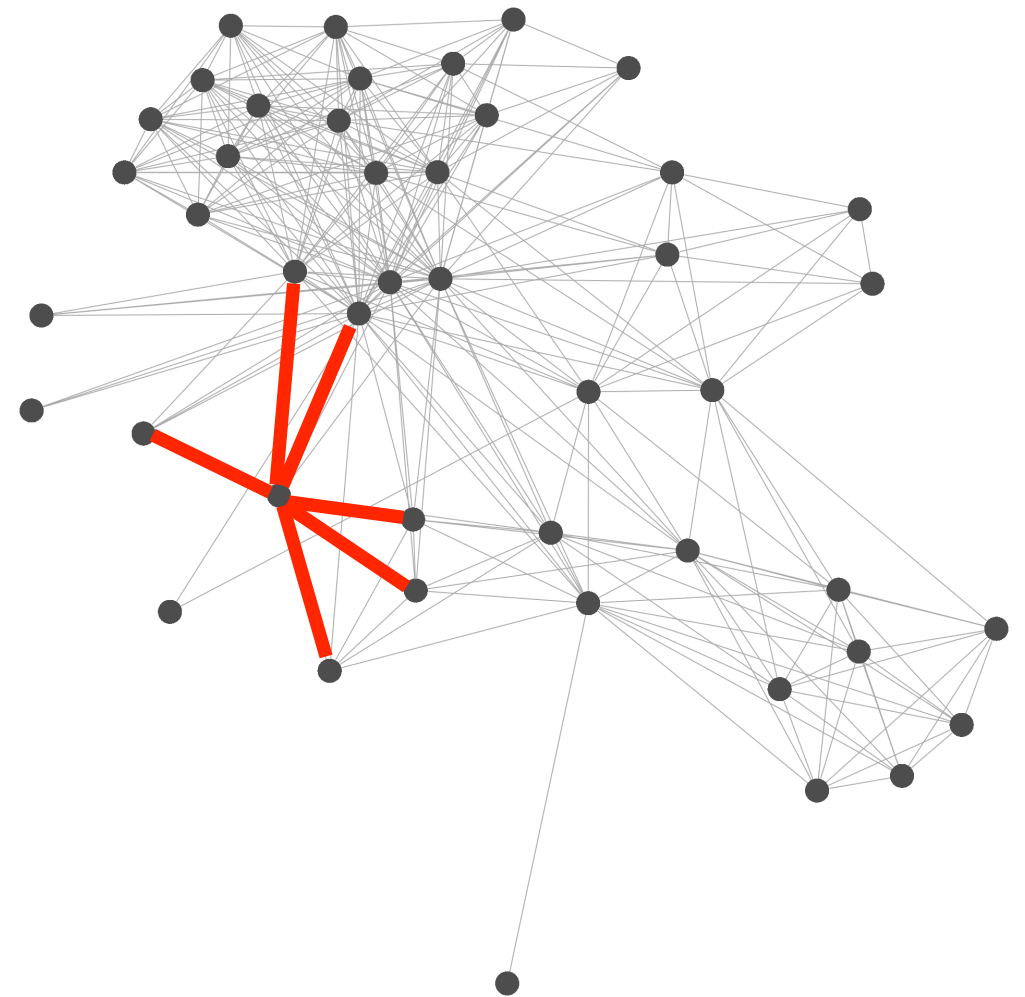
## Characteristics of ego-network structure



# Ego-network structural measures

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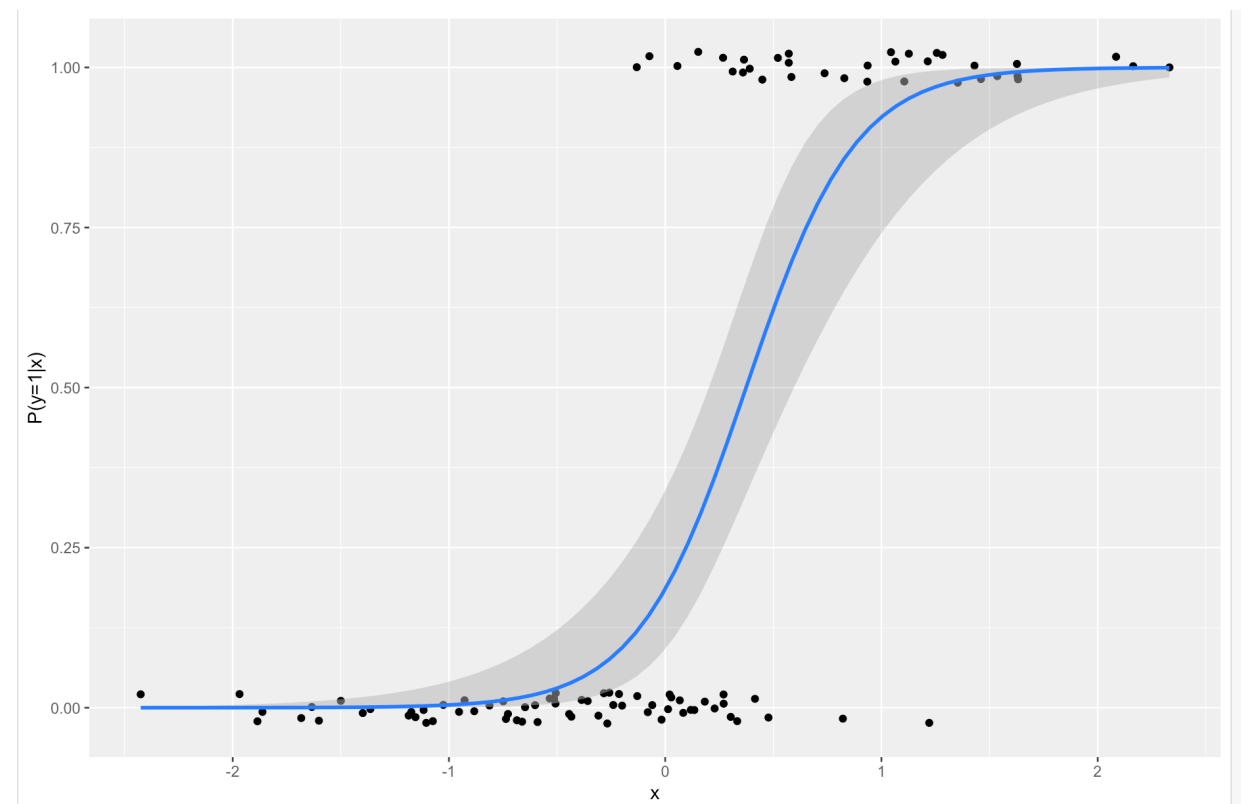
- Any SNA structural measure can be technically applied, but understanding of it must be adjusted to different **meaning and construction** (e.g. boundary specification) of egocentric vs sociocentric data
- E.g. degree centrality of student as alter in personal network of a classmate has very different meaning from degree centrality of student in school's sociocentric network (we don't observe whole network structure in school)



# What do we do with these summary measures?

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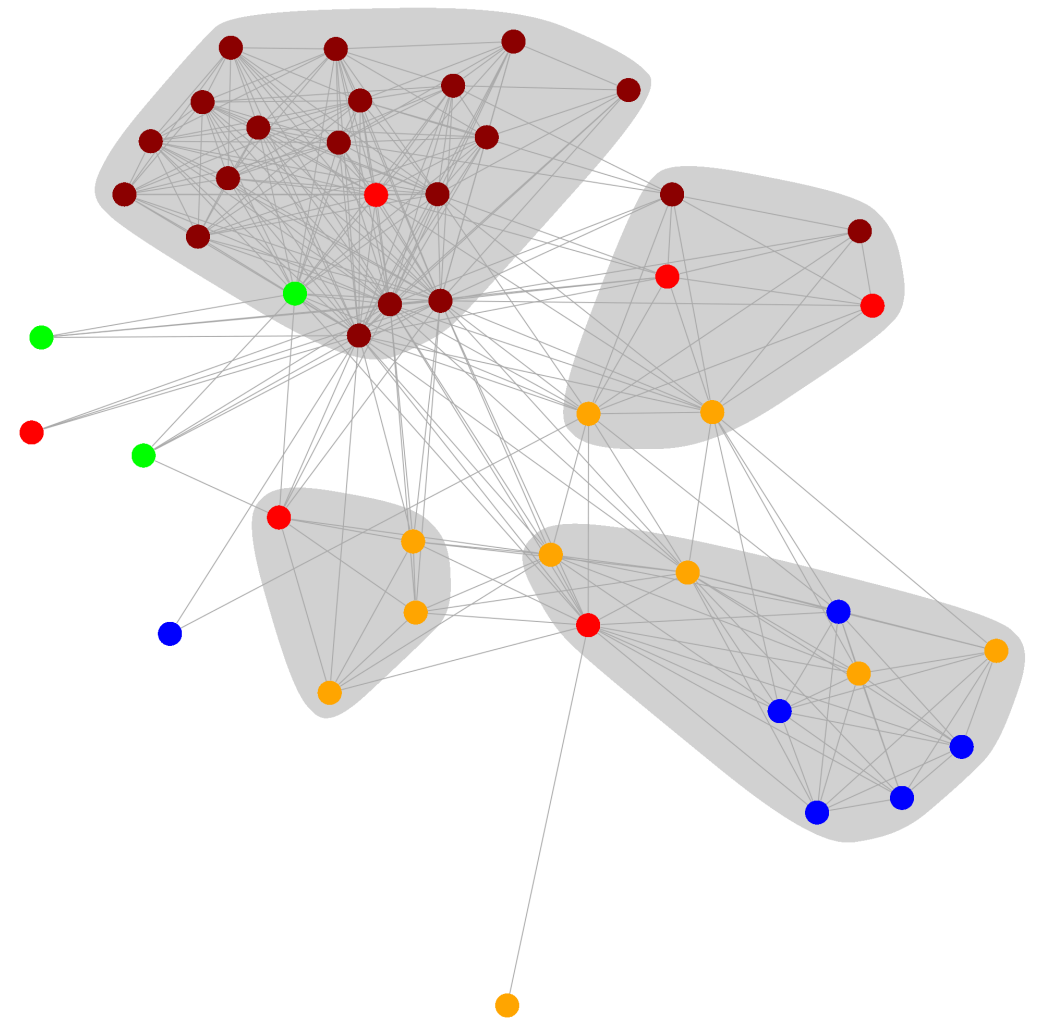
- Ego-level summary measures of network structure are used in the same **3 ways** as compositional measures
  - To describe their **univariate distribution**
  - To describe their **bivariate associations**
  - To estimate **statistical models**



# Combining compositional and structural information

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- A research question may require ego-network measures that combine information on alter attributes (**composition**) with information on distribution of alter-alter ties (**structure**)
- Examples:
  - Social context (family, work, school etc.) of alter with max betweenness
    - What is the compositional information?*
    - What is the structural information?*
  - Avg degree centrality of family alters
    - What is the compositional information?*
    - What is the structural information?*
  - *More examples?*

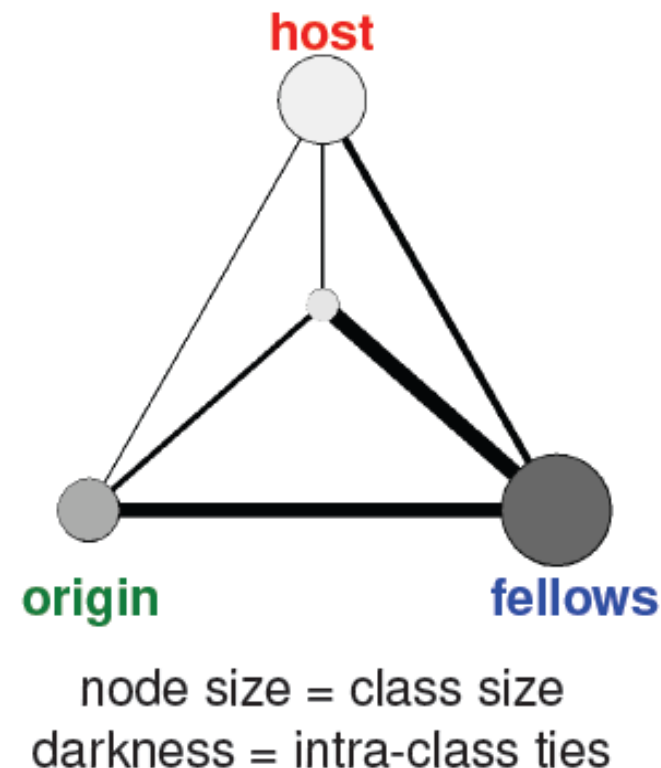
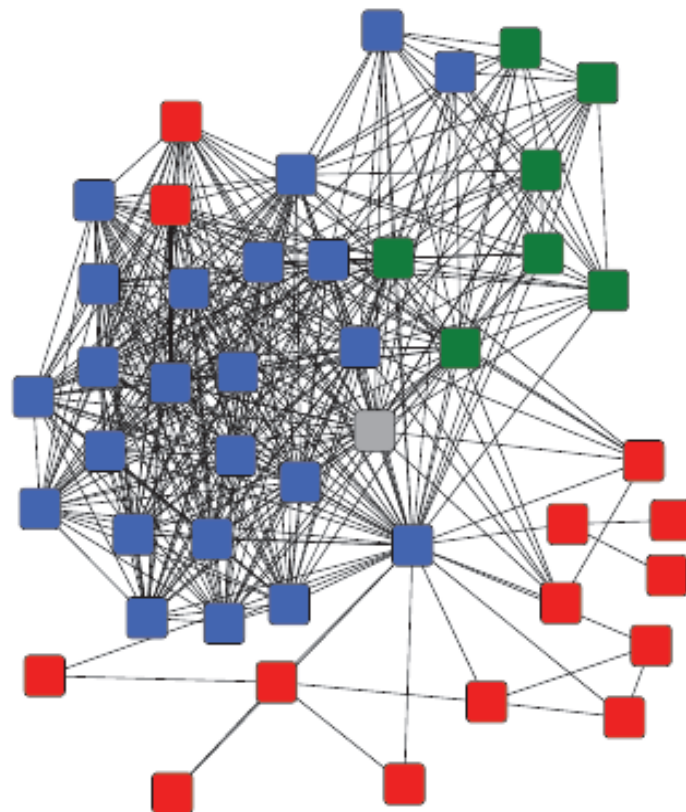


# Combining compositional and structural information

- **Clustered graph** with structural measures on density of connections between different compositional classes of alters in immigrants' personal networks (Brandes and Lerner)

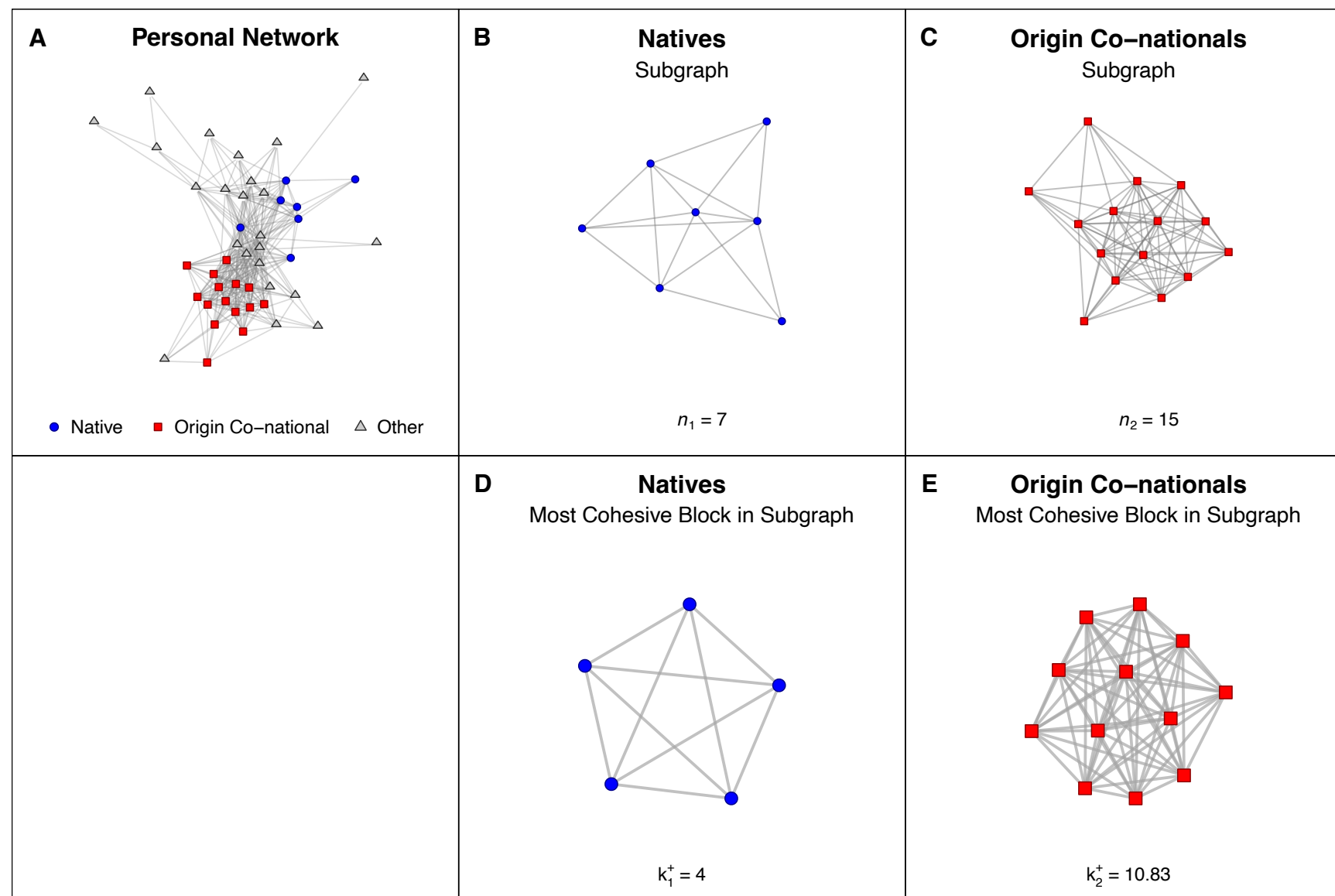
## Class-Level Network

argentinean woman living in spain



# Combining compositional and structural information

- Structural cohesion of different classes of alters in immigrants' personal networks



Vacca, R., Solano, G., Lubbers, M. J., Molina, J. L., & McCarty, C. (2018). A personal network approach to the study of immigrant structural assimilation and transnationalism. *Social Networks*.

# R time!



# Outline

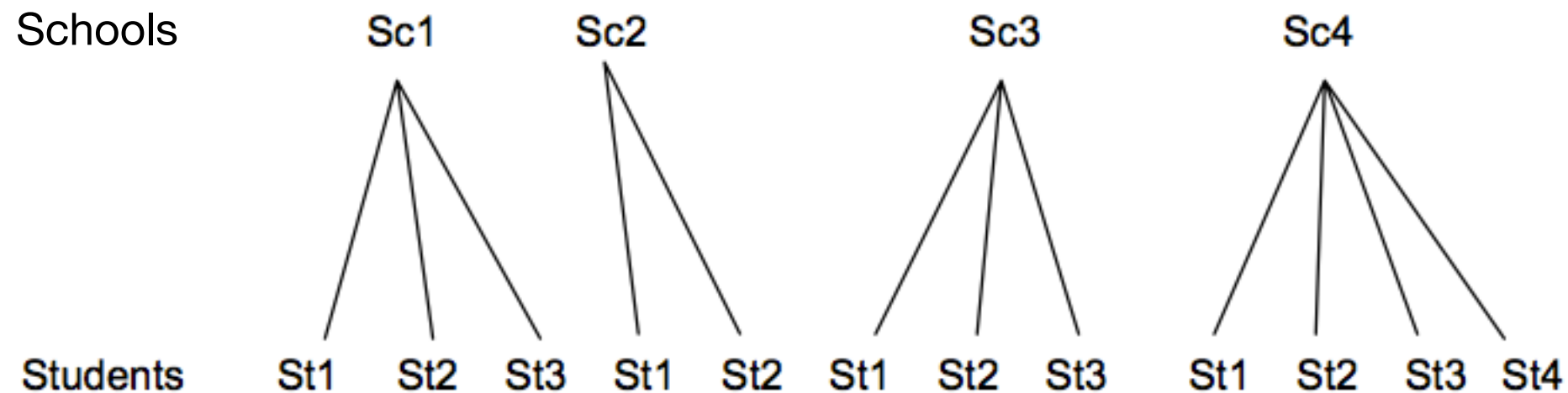
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1. The egocentric approach
2. Egocentric research design
3. Surveys on personal networks
4. Ego-network data
5. Ego-network composition
6. Ego-network structure
- 7. Multilevel models for egocentric networks**

# Multilevel data structures

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- Units of analysis are **clustered within groups**
- Examples:
  - Students within schools
  - Residents within neighborhoods
  - Individuals within countries
  - Patients within hospitals
  - **Ties or alters within ego-networks**

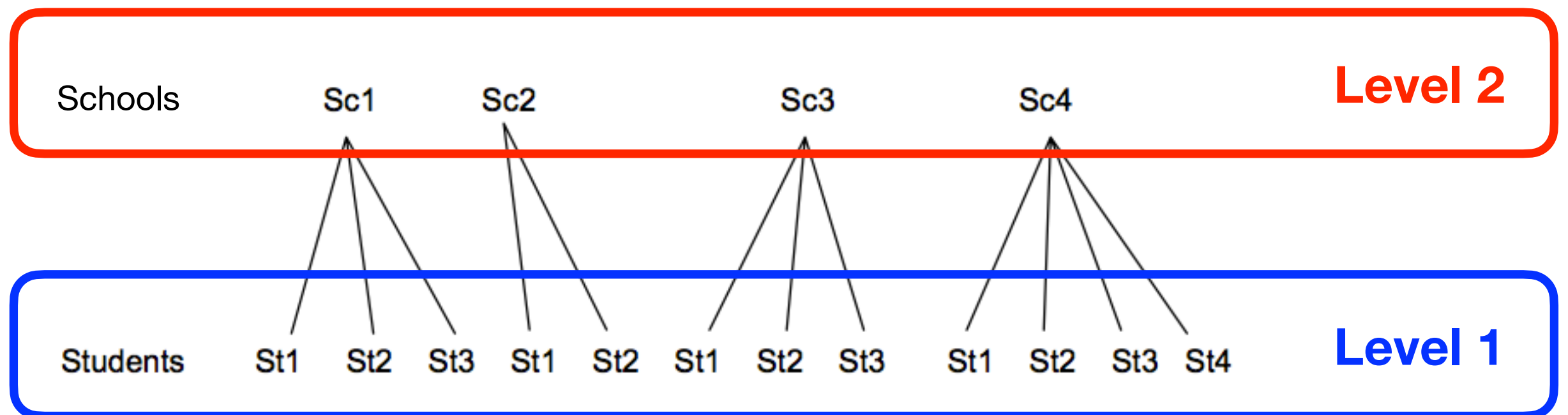


*(from Rasbash, Steele & Reckie, 2008)*

# Multilevel data structures

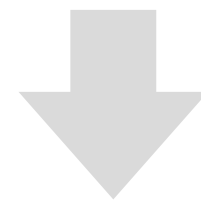
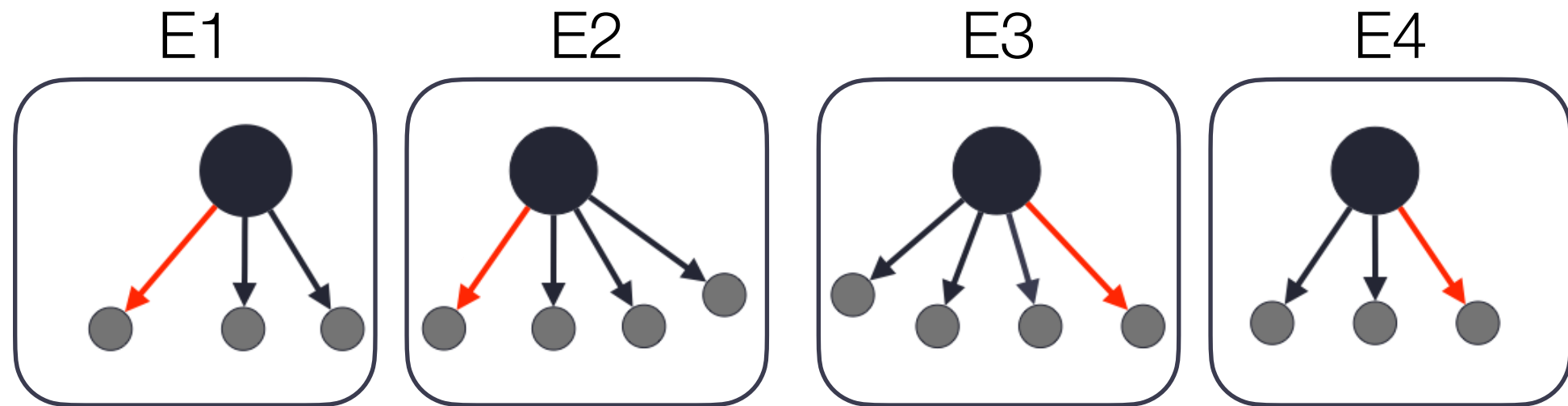
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- Examples:
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# Multilevel data structures: ego-network data

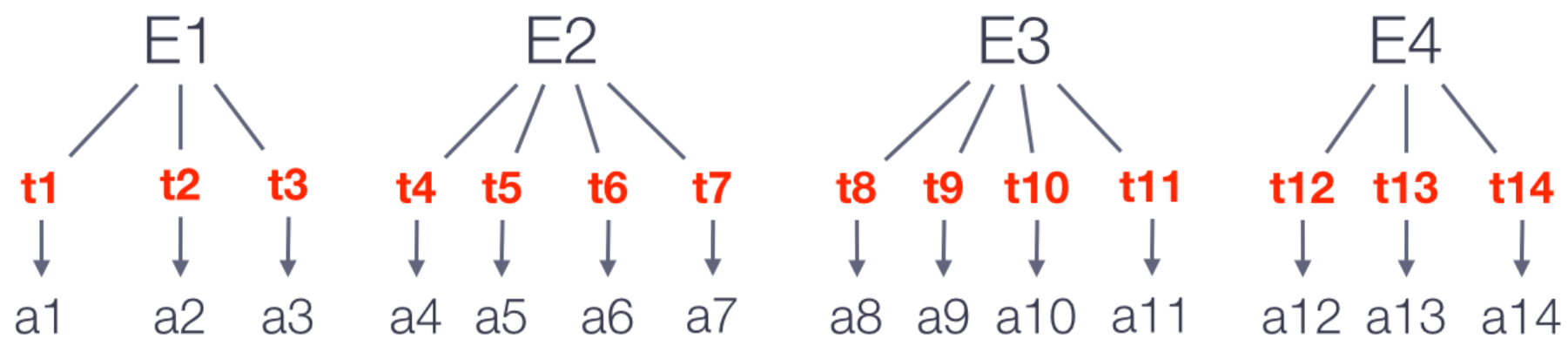
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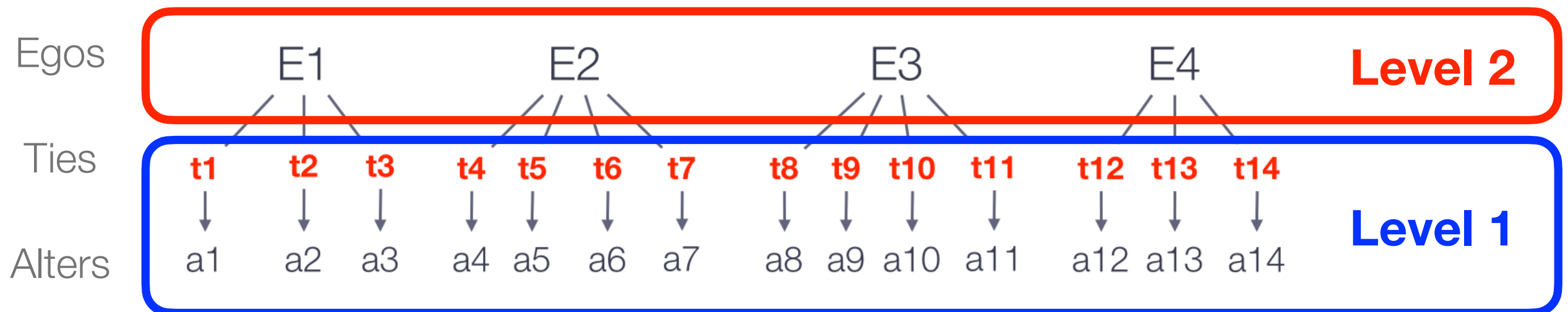
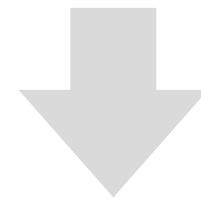
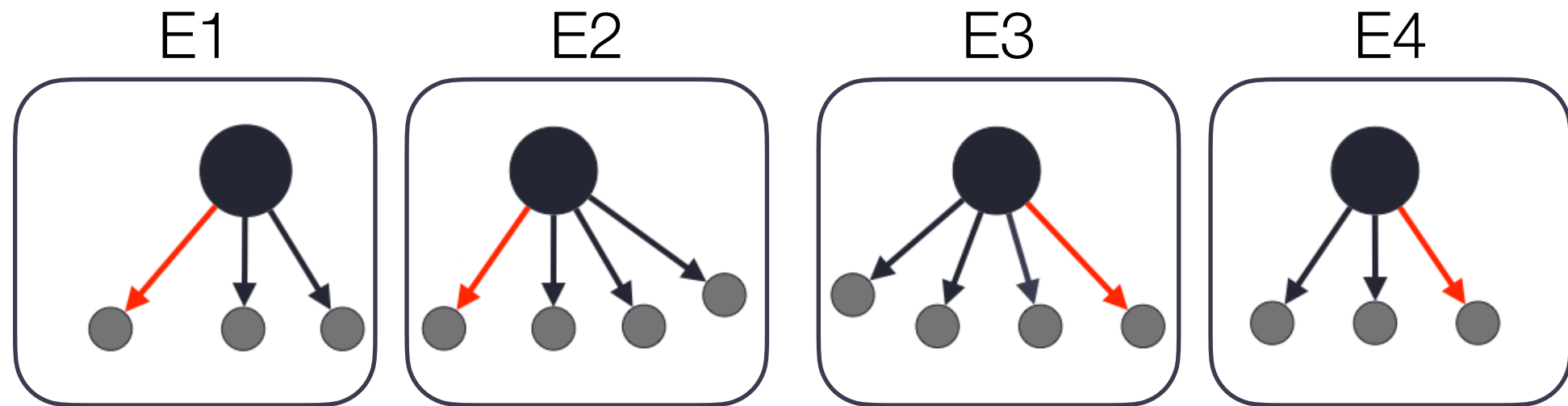
Egos

Ties

Alters

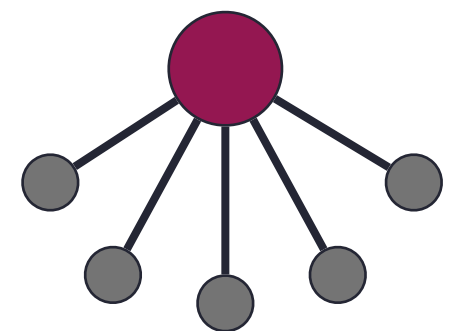
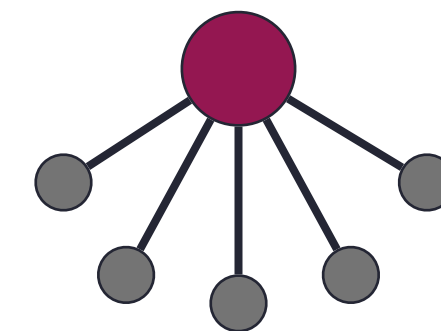
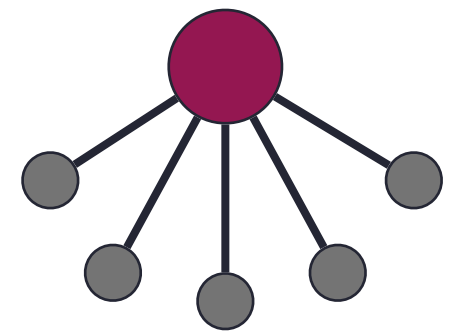
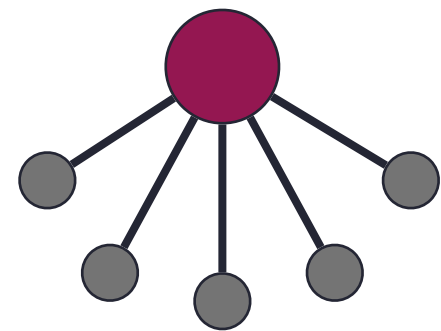
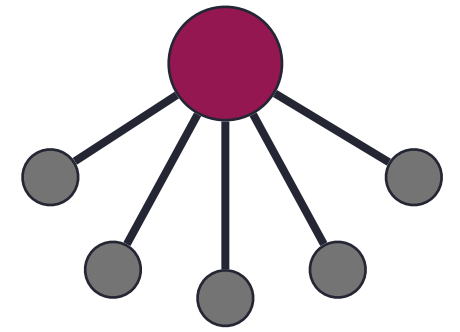
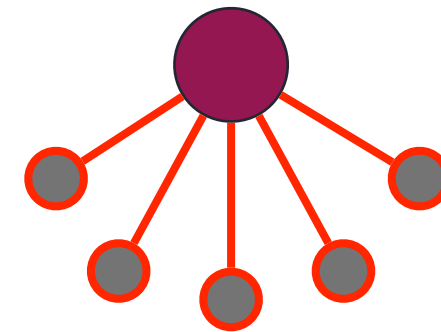


# Multilevel data structures: ego-network data



# Multilevel models for ego-network data

- In egocentric data, at least **two levels**
  - Level 1 (cases or “individuals”): Ego-alter ties or alters
  - Level 2 (“groups”): Ego-networks or egos
- In most egocentric data, ego-networks **do not overlap** (no ego-ego tie, no shared alter between egos)
- So each level-1 unit is **nested** in one and only one level-2 unit
- **Hierarchical** multilevel models can be used



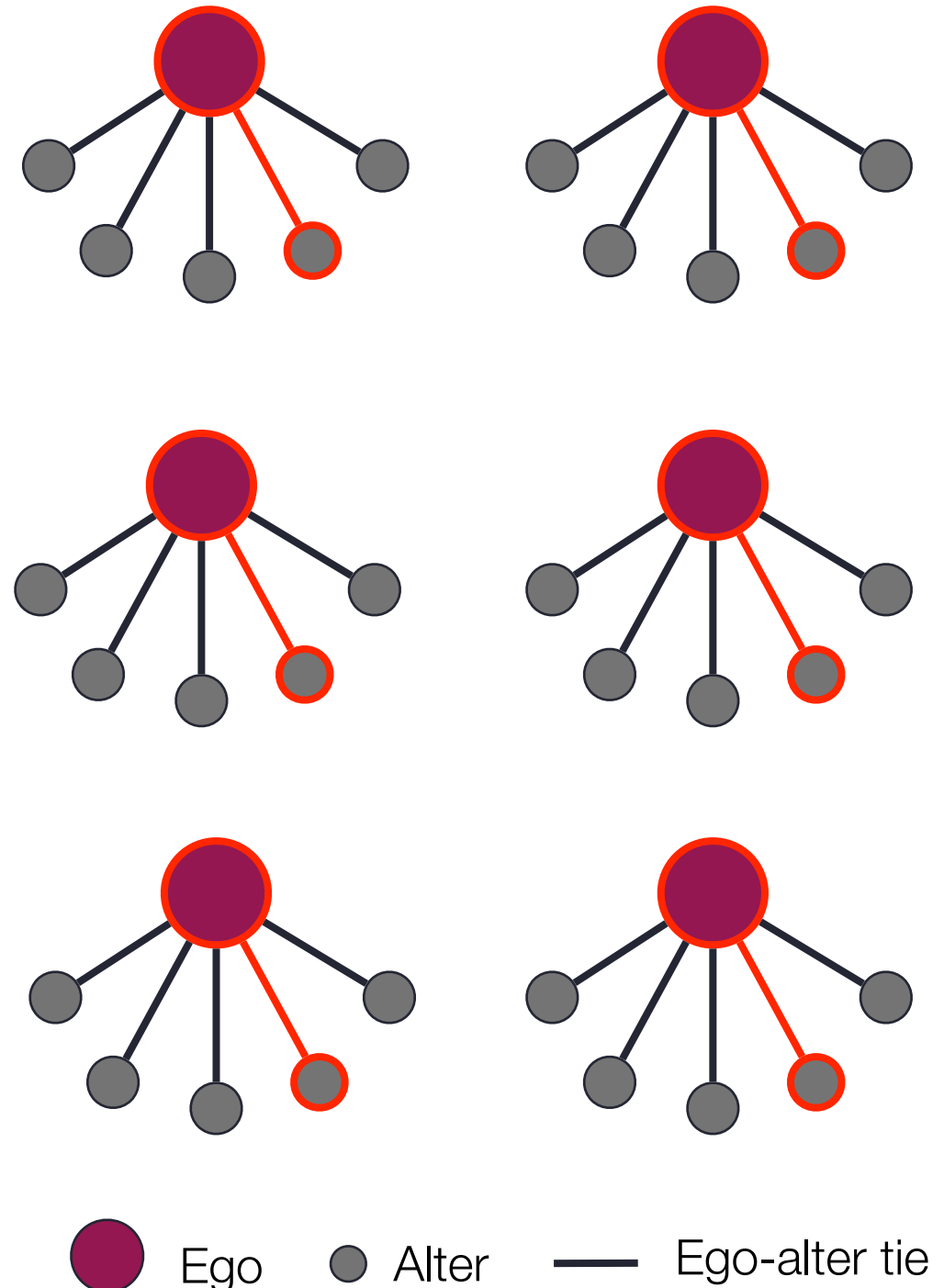
● Ego    ● Alter    — Ego-alter tie

# Multilevel models for ego-network data

- Dependent variable always measured at **level 1**: alter or tie

- **Examples:**

- Significance of cocaine use in ego-alter relationship (Snijders et al 1995):
  - ➔ **Continuous**
- Change in strength of ego-alter tie (van Duijn et al 1999)
  - ➔ **Continuous**
- Whether ego-alter tie provides support to ego (Wellman and Frank 2001, Herz 2015)
  - ➔ **Binary**
- Whether ego-alter tie of immigrant ego is to native alter (de Miguel Luken and Tranmer 2010)
  - ➔ **Binary**



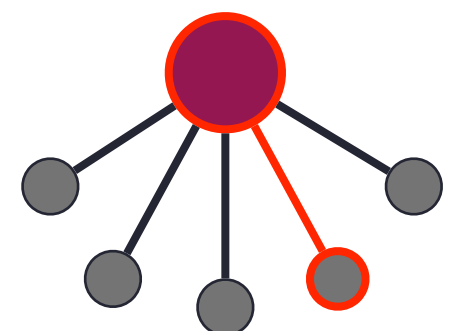
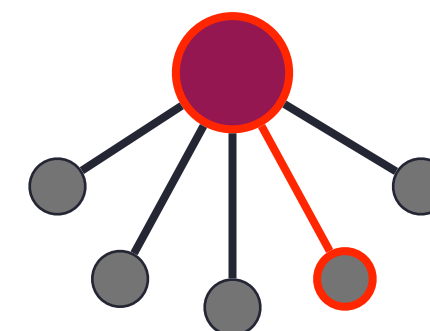
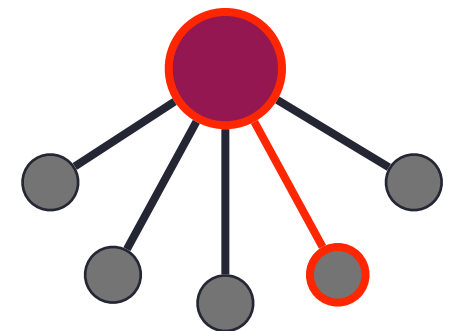
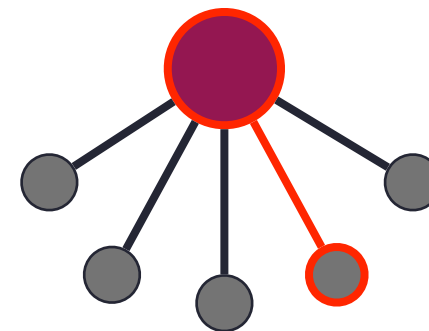
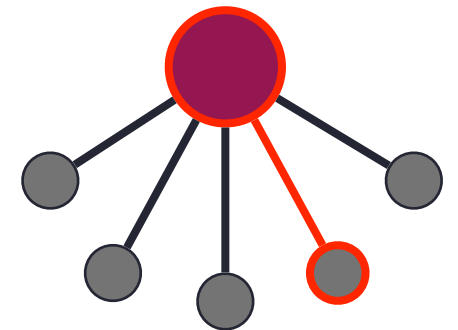
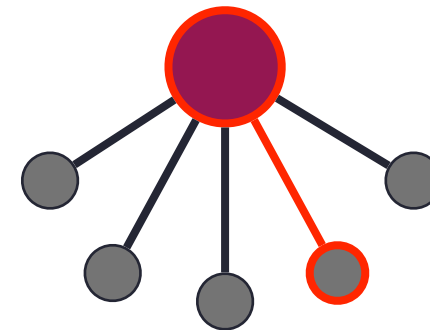
# Multilevel models for ego-network data

- **Linear** multilevel model if  $y$  is continuous:

$$y = \beta_0 + \beta_1 x + e$$

- **Logistic** multilevel model if  $y$  is binary:

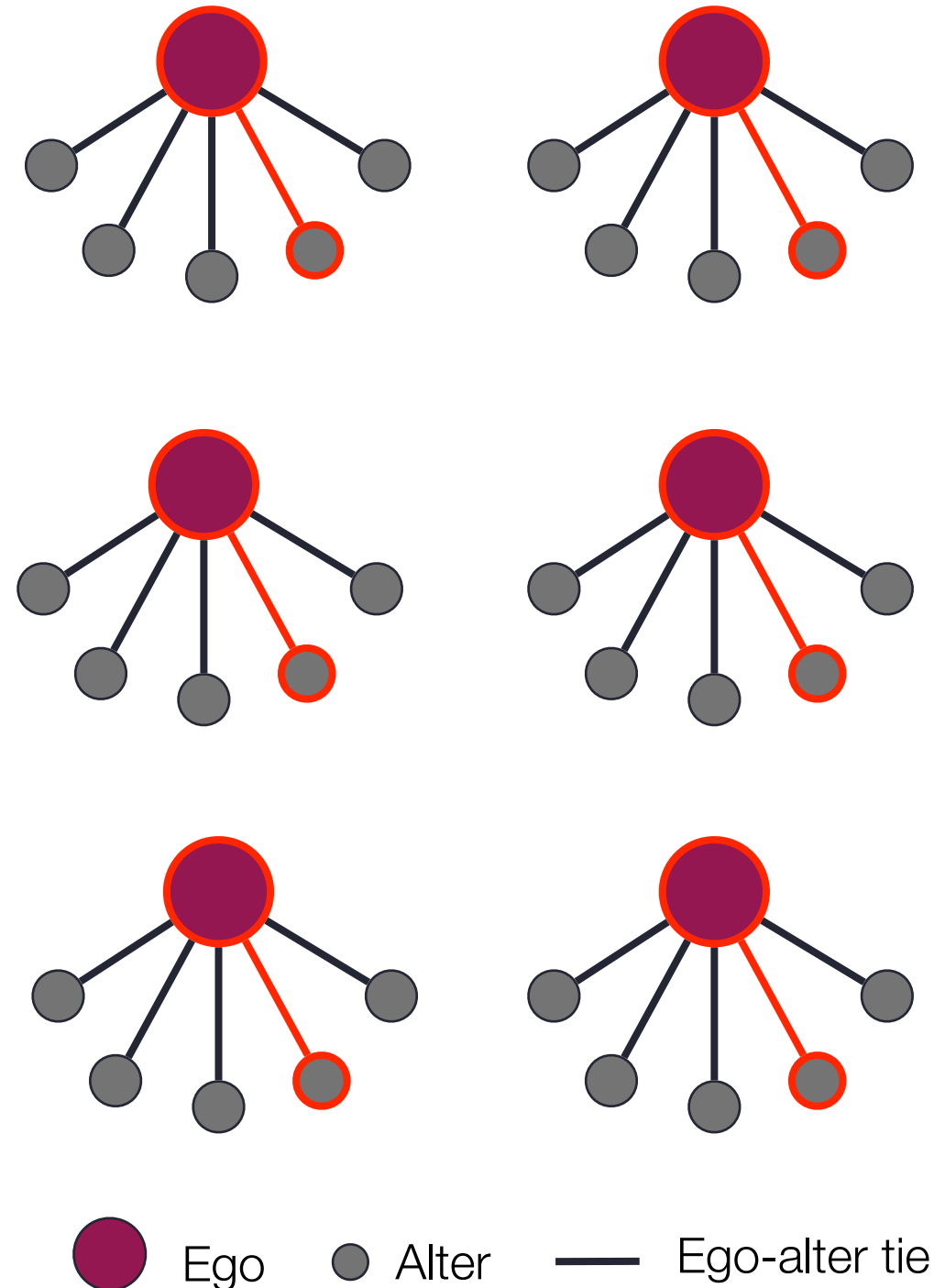
$$\log \left( \frac{\text{Prob}(y = 1)}{1 - \text{Prob}(y = 1)} \right) = \beta_0 + \beta_1 x + e$$



# Multilevel questions and hypotheses

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- Multilevel models can answer questions about both the **mean** and the **variance** of the DV
- And about how DV variation is generated at different levels (level 1, level 2, ...)



# Multilevel questions and hypotheses

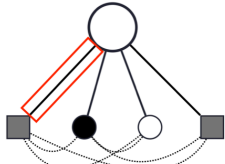
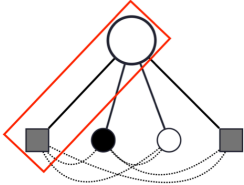
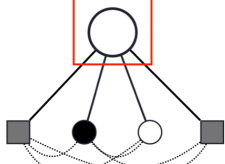
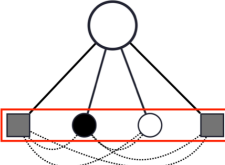
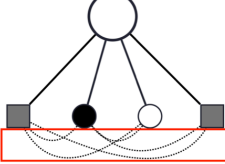
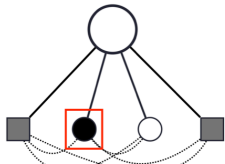
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- **Example:** model for support provided by ties (level 1) nested in egos (level 2):
  - Do ties with family tend to provide more support than ties with friends? (*mean*)
  - Is tendency of ties to give support more similar within the same ego-network than between different ego-networks? (*variance*)
  - What is the between-ego variation in tie supportiveness? (*variance*)
  - Is tie tendency to give support generally lower for ego A compared to ego B? (*mean*)
  - Does tie tendency to give support tend to be higher for older egos than for younger egos? (*mean*)
  - Does the effect of “being family” on tie supportiveness score (i.e. the average difference in support between family and non-family ties) change between egos? (*mean and variance*)
  - Is tie supportiveness more variable among female or male egos? (*variance*)

# Multilevel models for ego-networks: questions and variables

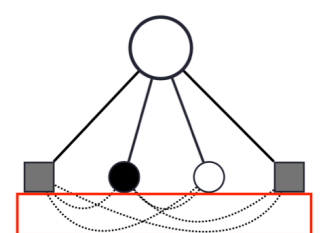
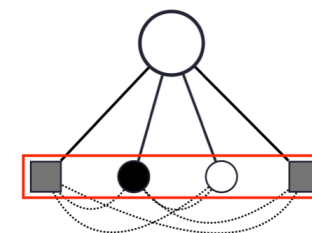
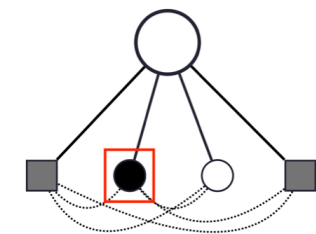
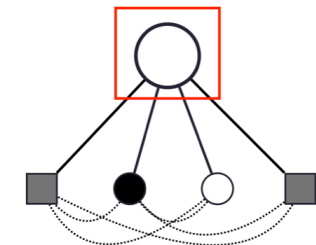
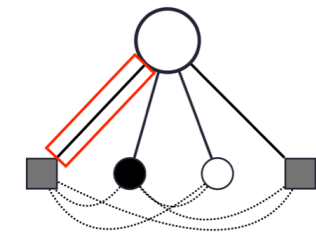
- Very flexible method to test all kind of hypotheses on how **outcome of ego-alter tie** (or alter) is influenced by

- Characteristics of **ego**
- Characteristics of **alter**
- Characteristics of ego-alter **tie**
- Characteristics of ego-alter **dyad**
- Characteristics of social context (**ego-network**)
- **Interactions** between these variables

|                                                                            | Measured unit | Level in hierarchical multilevel structure |                                                                                       |
|----------------------------------------------------------------------------|---------------|--------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Independent variables</b>                                               |               |                                            |                                                                                       |
| Characteristics of tie<br>(e.g. frequency of contact)                      | Tie           | 1                                          |    |
| Characteristics of ego-alter dyad<br>(e.g. age similarity)                 | Tie           | 1                                          |   |
| Characteristics of ego<br>(e.g. ego's age)                                 | Ego           | 2                                          |  |
| Characteristics of network composition<br>(e.g. proportion women)          | Ego-network   | 2                                          |  |
| Characteristics of network structure<br>(e.g. density of alter-alter ties) | Ego-network   | 2                                          |  |
| Characteristics of alter<br>(e.g. alter's age)                             | Alter         | 1                                          |  |

# Research question examples

- **Social support:** What social ties are more supportive among Sri Lankan immigrants in Italy?
- DV: Measure of support provided by ego-alter tie
- IVs:
  - **Characteristics of ego:** Do immigrants who are older tend to have more supportive ties?
  - **Characteristics of alter:** Do co-ethnic Sri Lankan contacts provide more support compared to non-ethnic (e.g. Italian native) contacts?
  - **Characteristics of ego-alter tie:** Are family ties more supportive than friendship ties?
  - **Characteristics of ego-alter dyad:** Are ties more supportive when ego and alter are of same sex, or similar age?
  - **Characteristics of social context** (ego-network): Are ties more supportive in social contexts in which family is more present (i.e. higher proportion family in ego-network)?
  - **Interactions between these variables:** Are family ties more supportive in personal networks with higher proportion of family?



# Social network surveys and software

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- A specific example: reproducible project for Vacca & Bianchi (2024)
  - [https://raffaelevacca.github.io/Vacca\\_Bianchi\\_2024\\_SSR/](https://raffaelevacca.github.io/Vacca_Bianchi_2024_SSR/)

